INDEX

Practical No - 1

Aim: Installing and setting environment variables for Working with Apache Hadoop

Hadoop Installation in Ubuntu

Introduction

Apache Hadoop is an open-source software framework used to store, manage and process large datasets for various big data computing applications running under clustered systems. It is Javabased and uses Hadoop Distributed File System (HDFS) to store its data and process data using MapReduce. In this article, you will learn how to install and configure Apache Hadoop on Ubuntu

Install Java

sudo apt install openjdk-8-jre

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

mohit@mohit-virtual-machine:~$ sudo apt install openjdk-8-jre
[sudo] password for mohit:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni openjdk-8-jre-headless
```

sudo apt install openjdk-8-jdk

```
Processing triggers for man-db (2.10.2-1) ...
mohit@mohit-virtual-machine:~$ sudo apt install openjdk-8-jdk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

Verify the installed version of Java.

java -version

```
Setting up libx11-dev:amd64 (2:1.7.5-1ubuntu0.3) ...
Setting up libxt-dev:amd64 (1:1.2.1-1) ...
mohit@mohit-virtual-machine:~$ java -version
openjdk version "1.8.0_452"
OpenJDK Runtime Environment (build 1.8.0_452-8u452-ga~us1-0ubuntu1~22.04-b09)
OpenJDK 64-Bit Server VM (build 25.452-b09, mixed mode)
```

Create Hadoop User and Configure Password-less SSH

Add a new user hadoop. sudo adduser hadoop

```
mohit@mohit-virtual-machine:~$ sudo adduser hadoop
Adding user `hadoop' ...
Adding new group `hadoop' (1001) ...
Adding new user `hadoop' (1001) with group `hadoop' ...
Creating home directory `/home/hadoop' ...
Copying files from `/etc/skel' ...
```

Add the hadoop user to the sudo group.

```
mohit@mohit-virtual-machine:~$ sudo usermod -aG sudo hadoop
mohit@mohit-virtual-machine:~$
```

Install the OpenSSH server and client.

```
mohit@mohit-virtual-machine:~$ sudo apt install openssh-server openssh-client -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

When you get a prompt, respond with: keep the local version currently installed
Log in with hadoop user. sudo su – hadoop

```
mohit@mohit-virtual-machine:~$ sudo su - hadoop
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

hadoop@mohit-virtual-machine:~$
```

Generate public and private key pairs. ssh-
keygen -t rsa
(don’t add password just press enter as many times to reach normal prompt)

```
hadoop@mohit-virtual-machine:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hadoop/.ssh/id_rsa):
Created directory '/home/hadoop/.ssh'.
Enter passphrase (empty for no passphrase):
```

Add the generated public key from id_rsa.pub to authorized_keys.
sudo cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys

```
+----[SHA256]-----+
hadoop@mohit-virtual-machine:~$ sudo cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_
keys
[sudo] password for hadoop:
hadoop@mohit-virtual-machine:~$
```

Change the permissions of the authorized_keys file.
sudo chmod 640 ~/.ssh/authorized_keys

```
hadoop@mohit-virtual-machine:~$ sudo chmod 640 ~/.ssh/authorized_keys
hadoop@mohit-virtual-machine:~$
```

Verify if the password-less SSH is functional.
ssh localhost

```
hadoop@mohit-virtual-machine:~$ ssh localhost
```

Install Apache Hadoop

Download the latest stable version of Hadoop. To get the latest version, go to Apache Hadoop official download page. wget https://archive.apache.org/dist/hadoop/core/hadoop-3.1.1/hadoop-3.1.1.tar.gz

```
hadoop@mohit-virtual-machine:~$ wget https://archive.apache.org/dist/hadoop/core
/hadoop-3.1.1/hadoop-3.1.1.tar.gz
--2025-06-10 09:52:49--  https://archive.apache.org/dist/hadoop/core/hadoop-3.1.
1/hadoop-3.1.1.tar.gz
```

Extract the downloaded file.
tar -xvzf hadoop-3.1.1.tar.gz

```
hadoop-3.1.1/share/doc/hadoop/hadoop-client-runtime/css/site.css
hadoop-3.1.1/share/doc/hadoop/hadoop-client-runtime/css/maven-theme.css
hadoop-3.1.1/share/doc/hadoop/hadoop-client-runtime/css/print.css
hadoop-3.1.1/share/doc/hadoop/hadoop-client-runtime/dependency-analysis.html
```

Move the extracted directory to the /usr/local/ directory.
sudo mv hadoop-3.1.1 /usr/local/hadoop

```
hadoop-3.1.1/share/doc/hadoop/hadoop-client-runtime/css/print.css
hadoop-3.1.1/share/doc/hadoop/hadoop-client-runtime/dependency-analysis.html
hadoop@mohit-virtual-machine:~$ sudo mv hadoop-3.1.1 /usr/local/hadoop
hadoop@mohit-virtual-machine:~$
```

Create directory to store system logs.
sudo mkdir /usr/local/hadoop/logs

```
hadoop@mohit-virtual-machine:~$ sudo mkdir /usr/local/hadoop/logs
hadoop@mohit-virtual-machine:~$
```

Change the ownership of the hadoop directory.
sudo chown -R hadoop:hadoop /usr/local/hadoop

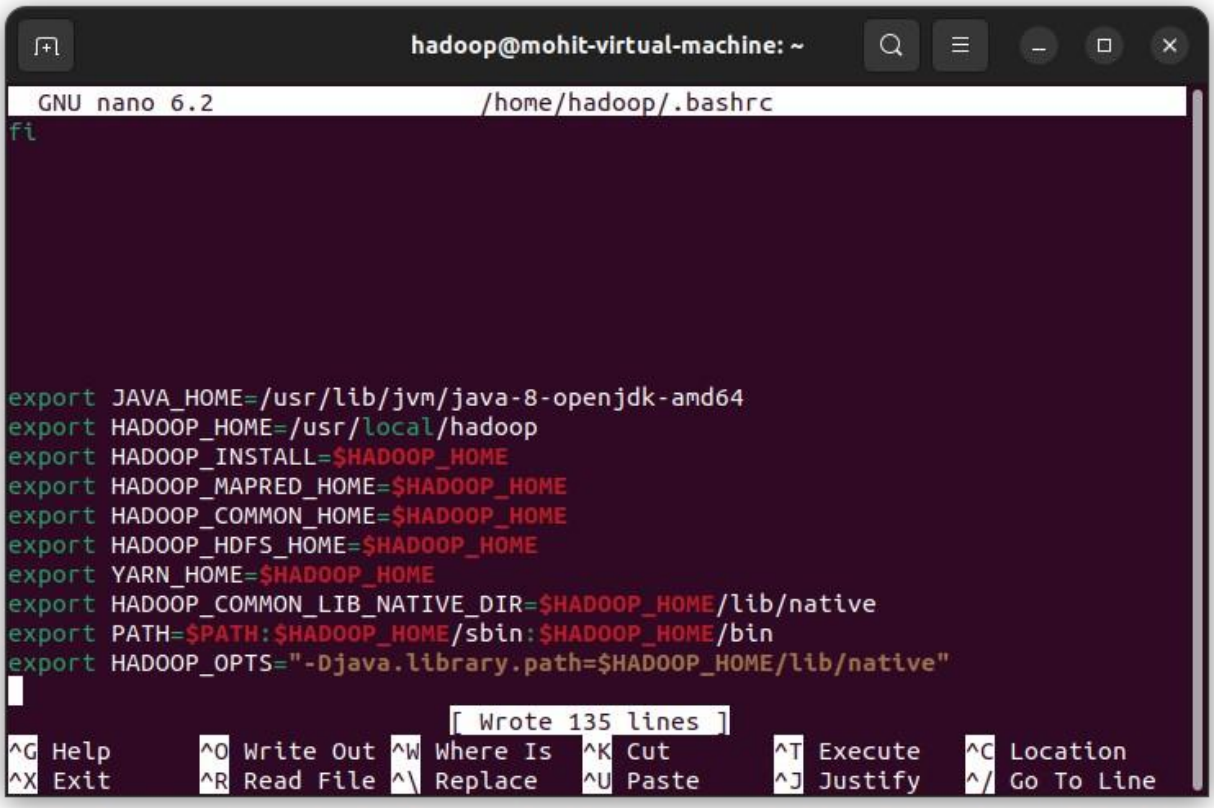
```
hadoop@mohit-virtual-machine:~$ sudo chown -R hadoop:hadoop /usr/local/hadoop
hadoop@mohit-virtual-machine:~$
```

Configure Hadoop

Edit file ~/.bashrc to configure the Hadoop environment variables.

```
hadoop@mohit-virtual-machine:~$ sudo nano ~/.bashrc
hadoop@mohit-virtual-machine:~$
```

Add the following lines to the file. Save and close the file.
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=/usr/local/hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export
HADOOP_MAPRED_HOME=$HADOOP_HOM
E export
HADOOP_COMMON_HOME=$HADOOP_HO
ME export
HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME export
HADOOP_COMMON_LIB_NATIVE_DIR=$HA
DOOP_HOME/lib/native export
PATH=$PATH:$HADOOP_HOME/sbin:$HADO
OP_HOME/bin export HADOOP_OPTS="-
Djava.library.path=$HADOOP_HOME/lib/native"



Activate the environment variables.
source ~/.bashrc

Configure Java Environment Variables

Hadoop has a lot of components that enable it to perform its core functions. To configure these components such as YARN, HDFS, MapReduce, and Hadoop-related project settings, you need to define Java environment variables in hadoop-env.sh configuration file.

Find the Java path.

which javac

```
hadoop@mohit-virtual-machine:~$ which javac
/usr/bin/javac
```

Find the OpenJDK directory.

readlink -f /usr/bin/javac

```
hadoop@mohit-virtual-machine:~$ readlink -f /usr/bin/javac
/usr/lib/jvm/java-8-openjdk-amd64/bin/javac
```

Edit the hadoop-env.sh file. sudo nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh

```
hadoop@mohit-virtual-machine:~$ sudo nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh
hadoop@mohit-virtual-machine:~$
```

Add the following lines to the file. Then, close and save the file.

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64 export HADOOP_CLASSPATH+=" $HAD

OOP_HOME/lib/*.jar"

Browse to the hadoop lib directory.
cd /usr/local/hadoop/lib

```
hadoop@mohit-virtual-machine:~$ cd /usr/local/hadoop/lib
hadoop@mohit-virtual-machine:/usr/local/hadoop/lib$
```

Download the Javax activation file. sudo wget
https://jcenter.bintray.com/javax/activation/javax.activation-api/1.2.0/javax.activationapi-1.2.0.jar

```
hadoop@mohit-virtual-machine:/usr/local/hadoop/lib$ sudo wget https://jcenter.b
intray.com/javax/activation/javax.activation-api/1.2.0/javax.activation-api-1.2
.0.jar
--2025-06-10 10:15:59--  https://jcenter.bintray.com/javax/activation/javax.act
ivation-api/1.2.0/javax.activation-api-1.2.0.jar
Resolving jcenter.bintray.com (jcenter.bintray.com)... 18.214.194.113, 18.232.1
72.199, 3.95.117.170
Connecting to jcenter.bintray.com (jcenter.bintray.com)|18.214.194.113|:443...
connected.
```

Come back to home folder of Hadoop user
cd /home/hadoop

```
hadoop@mohit-virtual-machine:/usr/local/hadoop/lib$ cd /home/hadoop
hadoop@mohit-virtual-machine:~$
```

Verify the Hadoop version.
hadoop version

```
hadoop@mohit-virtual-machine:~$ hadoop version
Hadoop 3.1.1
Source code repository https://github.com/apache/hadoop -r 2b9a8c1d3a2caf1e733d
57f346af3ff0d5ba529c
Compiled by leftnoteasy on 2018-08-02T04:26Z
Compiled with protoc 2.5.0
From source with checksum f76ac55e5b5ff0382a9f7df36a3ca5a0
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-
3.1.1.jar
hadoop@mohit-virtual-machine:~$
```

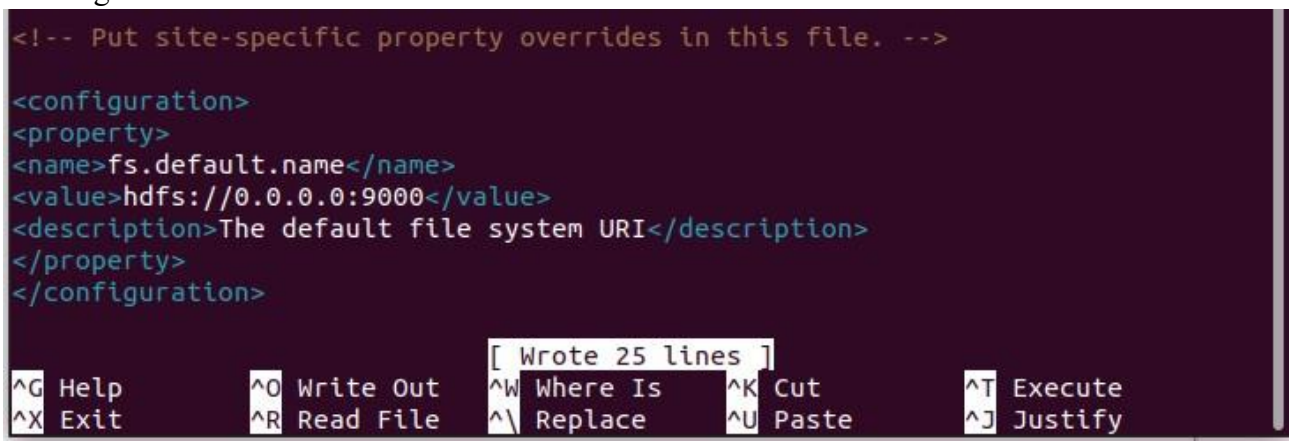
Edit the core-site.xml configuration file to specify the URL for your NameNode.
sudo nano $HADOOP_HOME/etc/hadoop/core-site.xml

```
hadoop@mohit-virtual-machine:~$ sudo nano $HADOOP_HOME/etc/hadoop/core-site.xml
hadoop@mohit-virtual-machine:~$
```

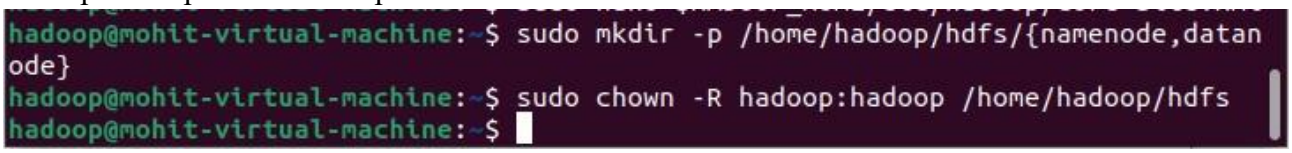
Add the following lines. Save and close the file.

```
<configuration>
<property>
<name>fs.default.name</name>
<value>hdfs://0.0.0.0:9000</value>
<description>The default file system URI</description>
```

</property>
</configuration>



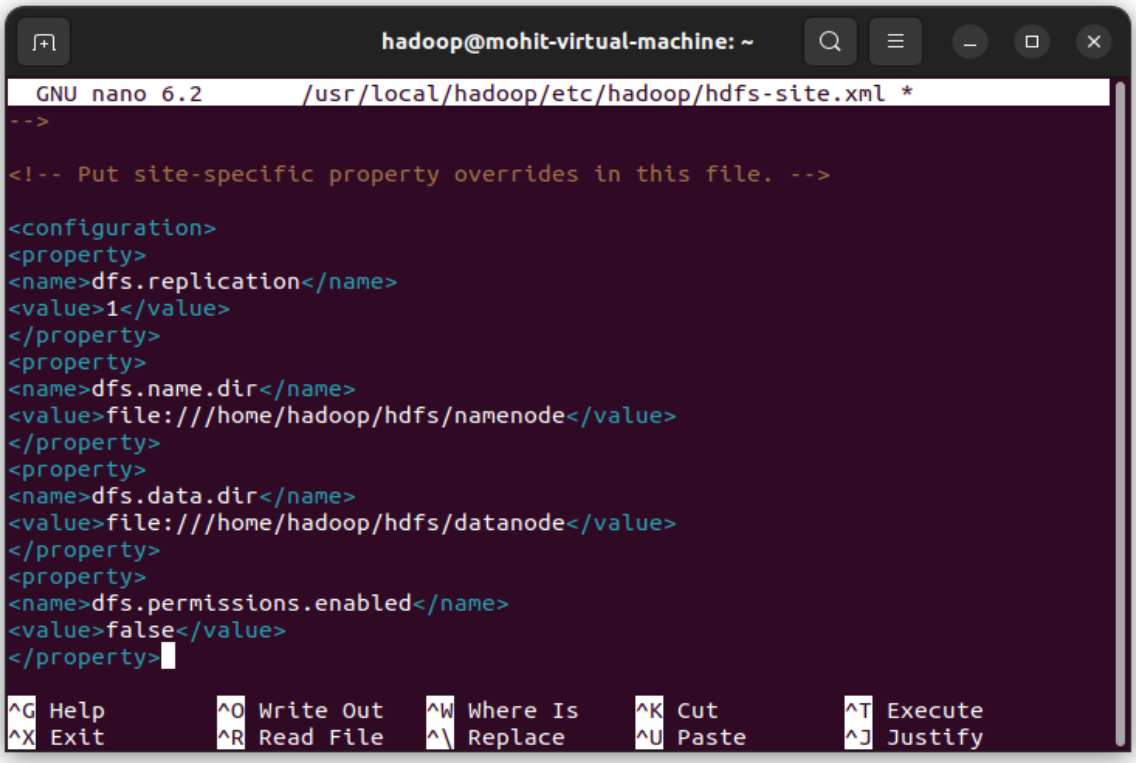
Create a directory for storing node metadata and change the ownership to hadoop.
sudo mkdir -p /home/hadoop/hdfs/{namenode,datanode} sudo chown -R
hadoop:hadoop /home/hadoop/hdfs



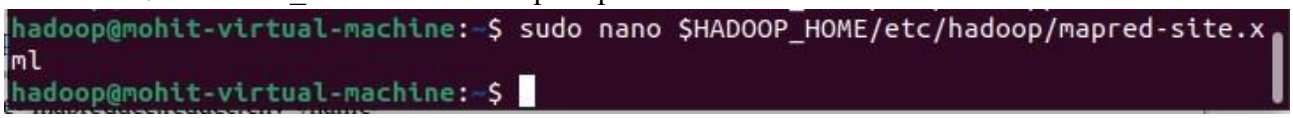
Edit hdfs-site.xml configuration file to define the location for storing node metadata, fs-image file.
sudo nano $HADOOP_HOME/etc/hadoop/hdfs-site.xml

Add the following lines. Close and save the file.
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.name.dir</name>
<value>file:///home/hadoop/hdfs/namenode</value>
</property>
<property>
<name>dfs.data.dir</name>
<value>file:///home/hadoop/hdfs/datanode</value>
</property>
<property>

<name>dfs.permissions.enabled</name>
<value>false</value>
</property>
</configuration>

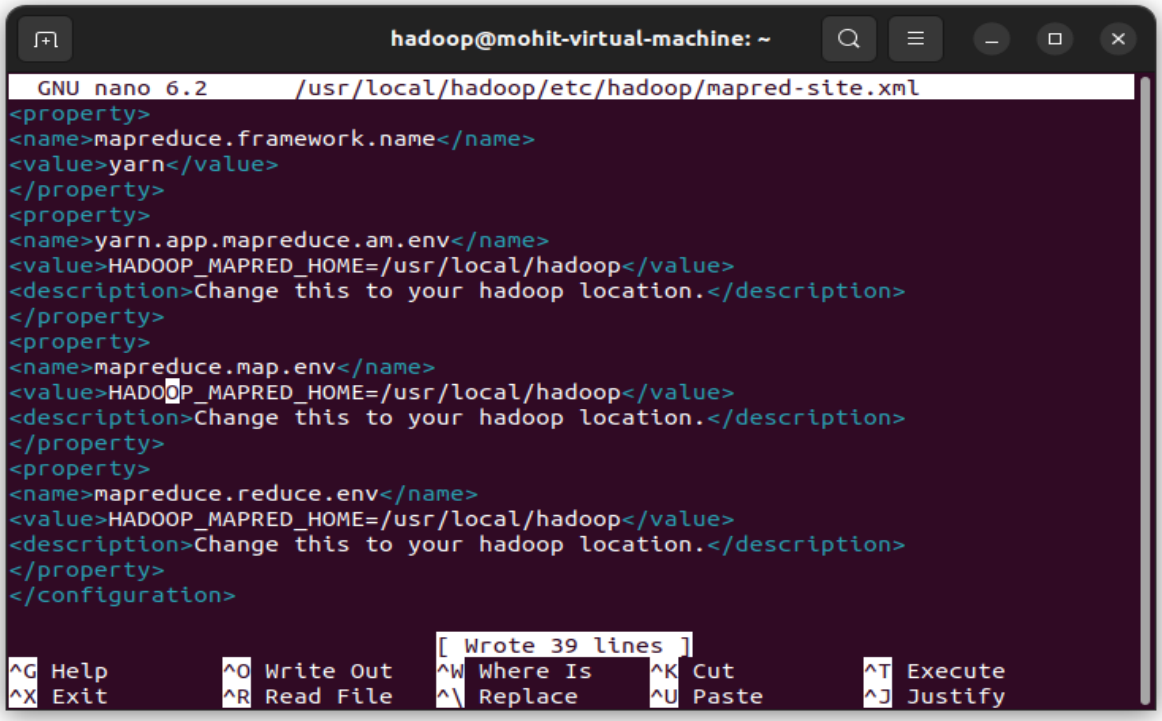


Edit mapred-site.xml configuration file to define MapReduce values.
sudo nano $HADOOP_HOME/etc/hadoop/mapred-site.xml

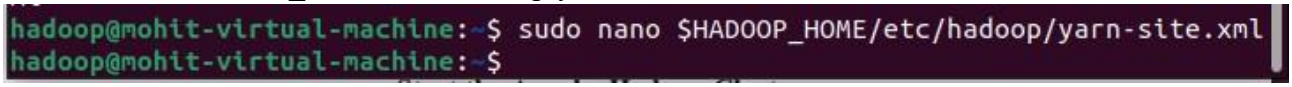


Add the following lines. Save and close the file.
<configuration>
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
<property>
<name>yarn.app.mapreduce.am.env</name>
<value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
<description>Change this to your hadoop location.</description>
</property>
<property>

<name>mapreduce.map.env</name>
<value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
<description>Change this to your hadoop location.</description>
</property>
<property>
<name>mapreduce.reduce.env</name>
<value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
<description>Change this to your hadoop location.</description>
</property>
</configuration>



Edit the yarn-site.xml configuration file and define YARN-related settings.
sudo nano $HADOOP_HOME/etc/hadoop/yarn-site.xml



Add the following lines. Save and close the file.
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>
</configuration>

```
-->
<configuration>

<!-- Site specific YARN configuration properties -->
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>
                              [ Wrote 22 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

Log in with hadoop user.(if you are not. This will be case once you have restarted you have computer) sudo su – hadoop

```
hadoop@mohit-virtual-machine:~$ sudo su - hadoop
hadoop@mohit-virtual-machine:~$
```

Validate the Hadoop configuration and format the HDFS NameNode.
hdfs namenode -format

```
hadoop@mohit-virtual-machine:~$ hdfs namenode -format
2025-06-10 10:42:12,193 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = mohit-virtual-machine/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.1.1
```

Start the Apache Hadoop Cluster

Start the NameNode and DataNode. start-dfs.sh

```
hadoop@mohit-virtual-machine:~$ start-dfs.sh
Starting namenodes on [0.0.0.0]
0.0.0.0: Warning: Permanently added '0.0.0.0' (ED25519) to the list of known ho
sts.
Starting datanodes
Starting secondary namenodes [mohit-virtual-machine]
mohit-virtual-machine: Warning: Permanently added 'mohit-virtual-machine' (ED25
519) to the list of known hosts.
hadoop@mohit-virtual-machine:~$
```

Start the YARN resource and node managers.
start-yarn.sh

```
hadoop@mohit-virtual-machine:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
hadoop@mohit-virtual-machine:~$
```

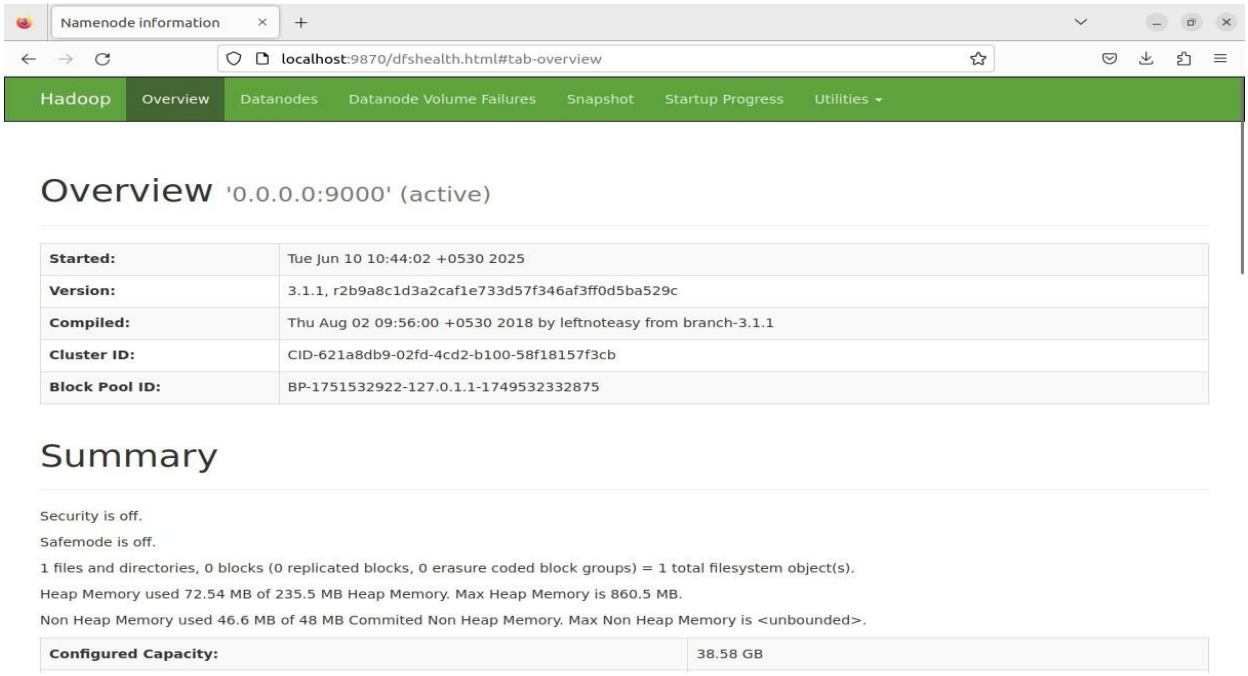
Verify all the running components.
Jps

```
hadoop@mohit-virtual-machine:~$ jps
12885 NodeManager
12533 ResourceManager
12021 DataNode
12204 SecondaryNameNode
11869 NameNode
13071 Jps
hadoop@mohit-virtual-machine:~$
```

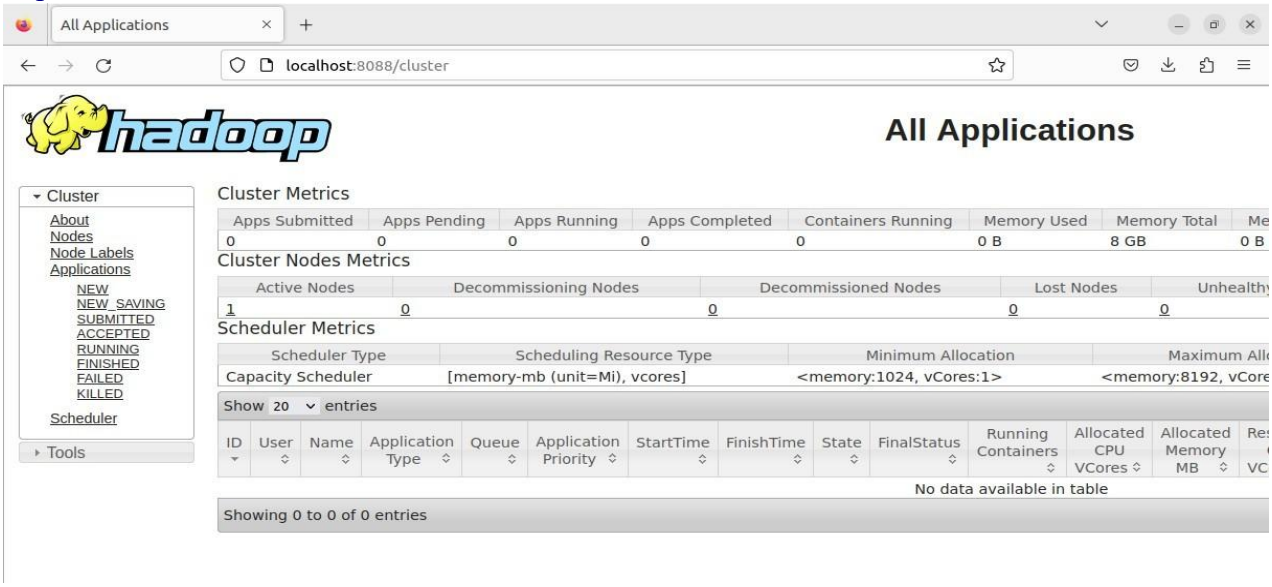
Access Apache Hadoop Web Interface

You can access the Hadoop NameNode and DataNode on your browser via http://localhost:9870. For example:

http://localhost:9870



http://localhost:8088



Practical No - 2

Aim: Implementing Map-Reduce Program for Word Count problem,

```
WC_Mapper.java package
com.wordcountproblem;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.LongWritable; import
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapred.MapReduceBase; import
org.apache.hadoop.mapred.Mapper; import
org.apache.hadoop.mapred.OutputCollector; import
org.apache.hadoop.mapred.Reporter;

/**
* WC_Mapper is the Mapper class for the Word Count problem.
* It extends MapReduceBase and implements the Mapper interface.
 *
* The map method takes a line of text as input, tokenizes it into words, * and emits each word along
  with a count of 1.
 */
public class WC_Mapper extends MapReduceBase implements
Mapper<LongWritable, Text, Text, IntWritable> {

   // A constant IntWritable with value 1, used for counting each word occurrence.
private final static IntWritable one = new IntWritable(1);     // A Text object to
hold the current word being processed.     private Text word = new Text();

   /**
* The map method processes a single line of input text.
    *
* @param key The byte offset of the current line in the input file (not used in this mapper).
* @param value The entire line of text as a Text object.
* @param output An OutputCollector to emit key-value pairs (word, 1).
* @param reporter A Reporter object for reporting progress and counters (not used in this mapper).
* @throws IOException If an I/O error occurs.
    */
   public void map(LongWritable key, Text value,
                   OutputCollector<Text, IntWritable> output,
```

```
            Reporter reporter) throws IOException {

      // Convert the input Text value to a Java String.
String line = value.toString();

      // Create a StringTokenizer to break the line into words.
      // By default, StringTokenizer uses whitespace as delimiters.
StringTokenizer tokenizer = new StringTokenizer(line);

      // Iterate through each token (word) in the line.
      while (tokenizer.hasMoreTokens()) {
         // Set the 'word' Text object to the current token.
word.set(tokenizer.nextToken());
         // Emit the word and a count of 1 to the OutputCollector.
output.collect(word, one);
      }
   }
}
```

WC_Reducer.java package
com.wordcountproblem;

```
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapred.MapReduceBase; import
org.apache.hadoop.mapred.OutputCollector; import
org.apache.hadoop.mapred.Reducer; import
org.apache.hadoop.mapred.Reporter;

/**
* WC_Reducer is the Reducer class for the Word Count problem.
* It extends MapReduceBase and implements the Reducer interface.
 *
* The reduce method takes a word as a key and an iterator of counts (1s) * as values. It sums these
  counts to get the total frequency of the word  * and emits the word along with its total count.
 */
public class WC_Reducer extends MapReduceBase implements
Reducer<Text, IntWritable, Text, IntWritable> {

   /**
* The reduce method aggregates the counts for a given word.
```

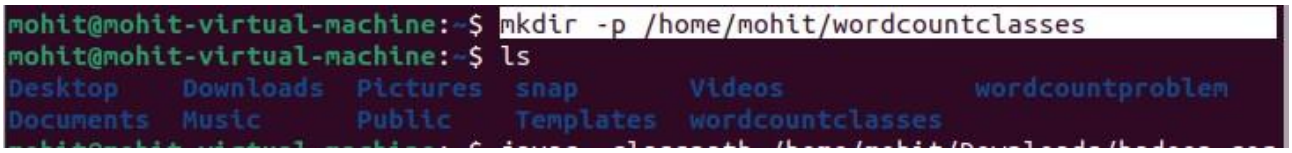
```
    *
* @param key The word (Text) whose counts are to be summed.
* @param values An Iterator of IntWritable objects, each representing a count of 1     * for the
  given word from the mappers.
* @param output An OutputCollector to emit the final word and its total count.
* @param reporter A Reporter object for reporting progress and counters (not used in this reducer).
* @throws IOException If an I/O error occurs.
    */
   public void reduce(Text key, Iterator<IntWritable> values,
                      OutputCollector<Text, IntWritable> output,
Reporter reporter) throws IOException {

      // Initialize a sum variable to accumulate the counts for the current word.
int sum = 0;

      // Iterate through all the IntWritable values (which are all '1's from the mappers)
// associated with the current word (key).        while (values.hasNext()) {
         // Get the integer value from the current IntWritable and add it to the sum.
sum += values.next().get();
      }

      // After summing all the counts for the word, emit the word (key)
// and its total count (sum) as a new IntWritable object.
output.collect(key, new IntWritable(sum));
   }
}
```

WC_Runner.java package com.wordcountproblem;
import java.io.IOException; import
org.apache.hadoop.fs.Path; import
org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapred.FileInputFormat; import
org.apache.hadoop.mapred.FileOutputFormat; import
org.apache.hadoop.mapred.JobClient; import
org.apache.hadoop.mapred.JobConf; import
org.apache.hadoop.mapred.TextInputFormat; import
org.apache.hadoop.mapred.TextOutputFormat; public
class WC_Runner {
public static void main(String[] args) throws IOException{
JobConf conf = new JobConf(WC_Runner.class);
conf.setJobName("WordCount");
conf.setOutputKeyClass(Text.class);

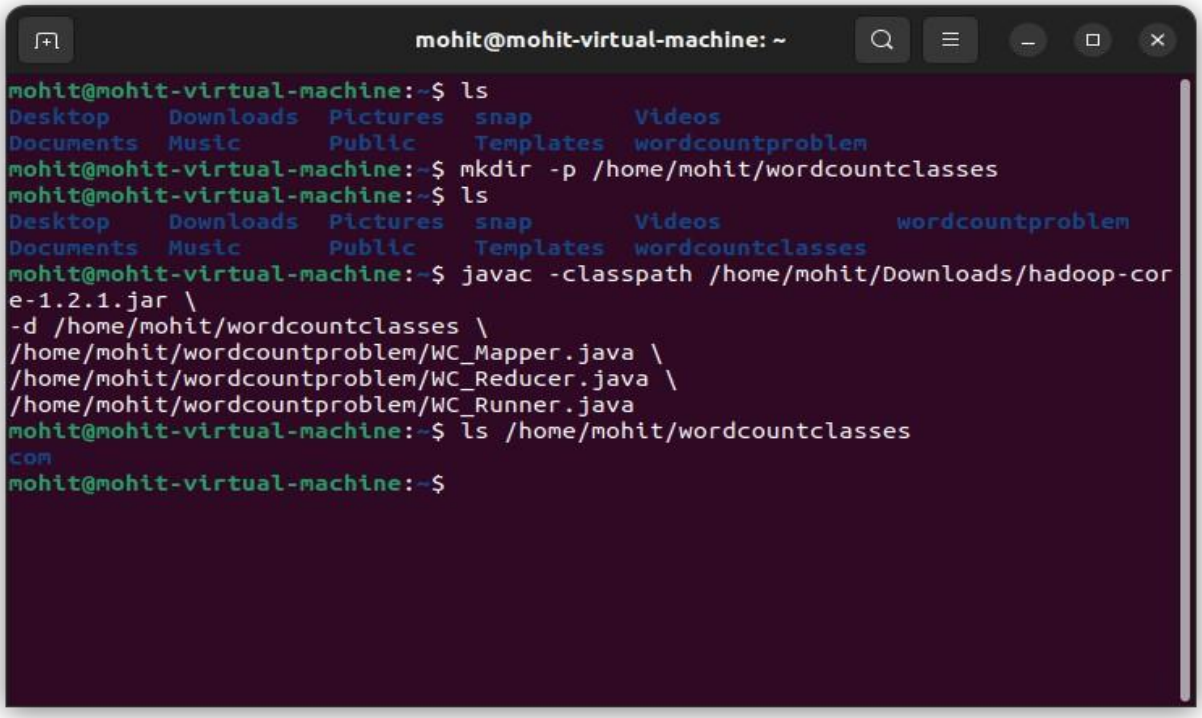
```
conf.setOutputValueClass(IntWritable.class);
conf.setMapperClass(WC_Mapper.class);
conf.setCombinerClass(WC_Reducer.class);
conf.setReducerClass(WC_Reducer.class);
conf.setInputFormat(TextInputFormat.class);
conf.setOutputFormat(TextOutputFormat.class);
FileInputFormat.setInputPaths(conf,new Path(args[0]));
FileOutputFormat.setOutputPath(conf,new Path(args[1]));
JobClient.runJob(conf);
}
}
```

Compiling and making jar files (make sure to login as root user)

mkdir -p /home/mohit/wordcountclasses




```
javac -classpath /home/mohit/Downloads/hadoop-core-1.2.1.jar \
-d /home/mohit/wordcountclasses \
/home/mohit/wordcountproblem/WC_Mapper.java \
/home/mohit/wordcountproblem/WC_Reducer.java \
/home/mohit/wordcountproblem/WC_Runner.java
```



cd /home/mohit/wordcountclasses jar -cvf

/home/mohit/wordcountproblem.jar com/

```
mohit@mohit-virtual-machine:~$ cd /home/mohit/wordcountclasses

jar -cvf /home/mohit/wordcountproblem.jar com/
added manifest
adding: com/(in = 0) (out= 0)(stored 0%)
adding: com/wordcountproblem/(in = 0) (out= 0)(stored 0%)
adding: com/wordcountproblem/WC_Runner.class(in = 1503) (out= 730)(deflated 51%)
adding: com/wordcountproblem/WC_Mapper.class(in = 1883) (out= 777)(deflated 58%)
adding: com/wordcountproblem/WC_Reducer.class(in = 1552) (out= 621)(deflated 59%
)
```

nano *home*/mohit/data.txt

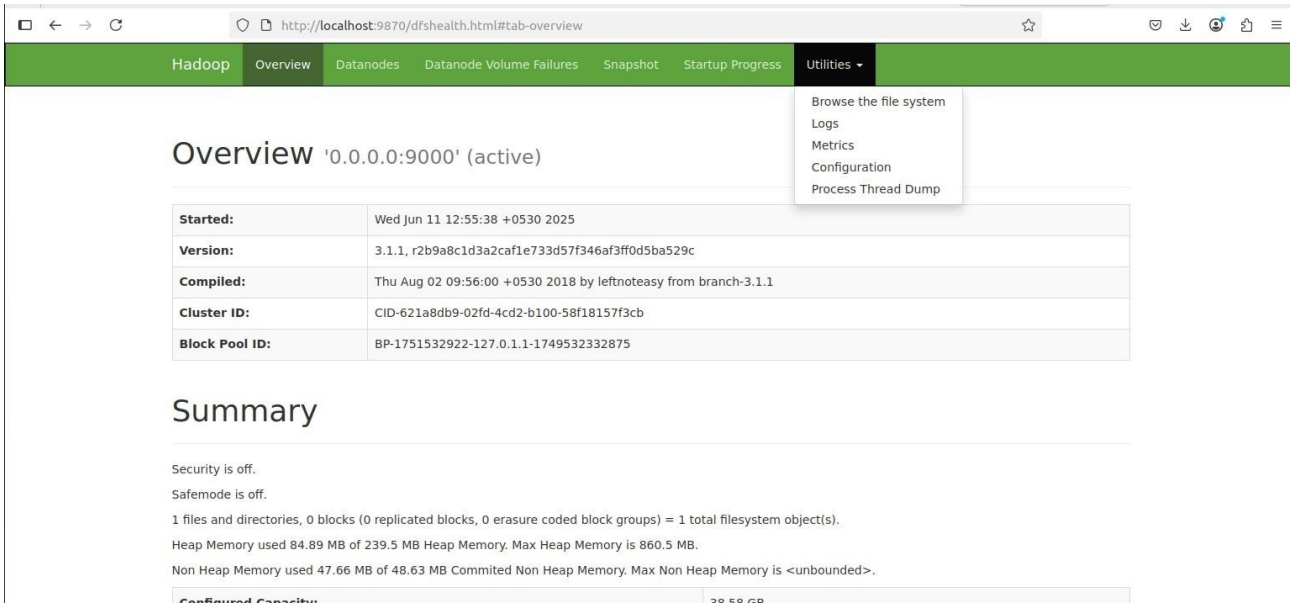
```
mohit@mohit-virtual-machine:~/wordcountclasses$ nano /home/mohit/data.txt
```

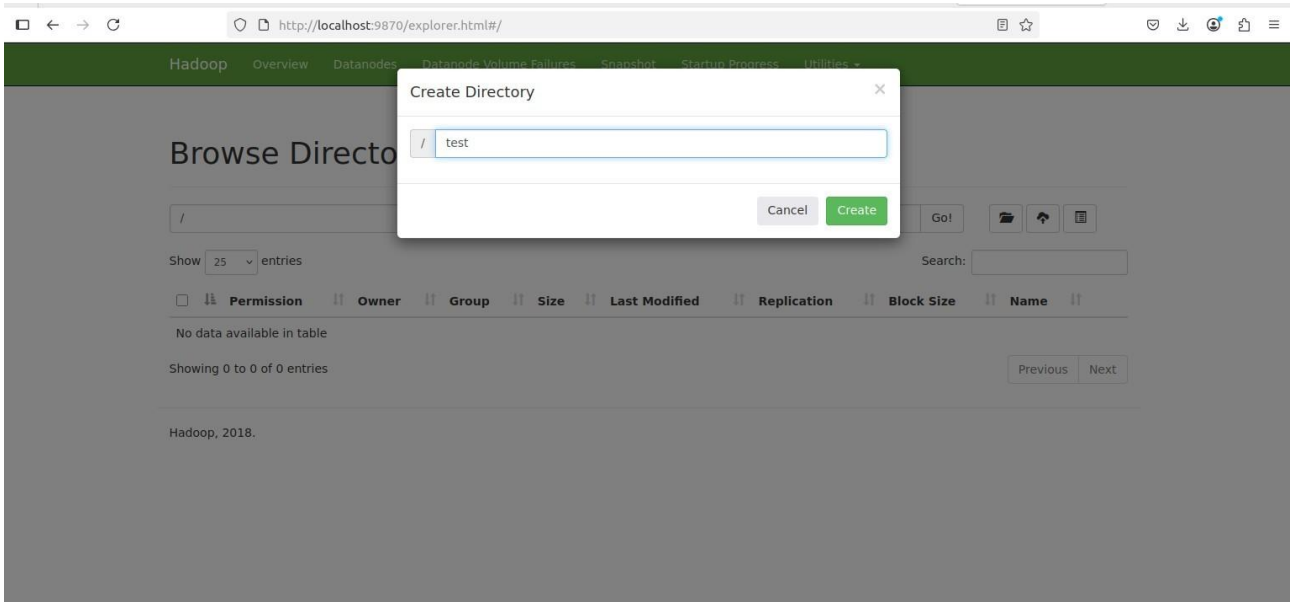
```
mohit@mohit-virtual-machine: ~/wordcountclasses
  GNU nano 6.2                    /home/mohit/data.txt
HDFS is a storage unit of Hadoop
MapReduce is a processing unit of hadoop

                        [ Read 2 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

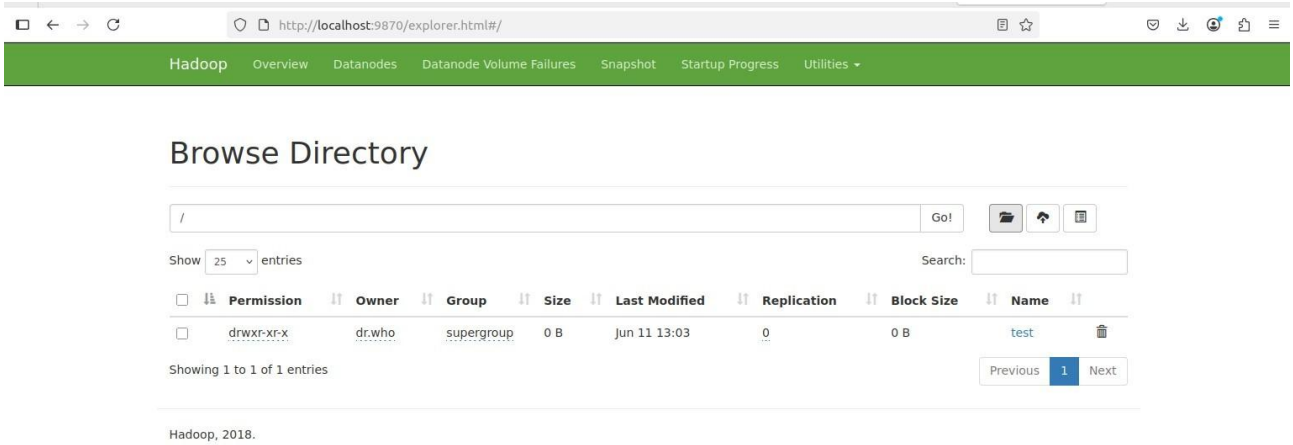
Start Hadoop server and yarn server (start-dfs.sh and start-yarn.sh)

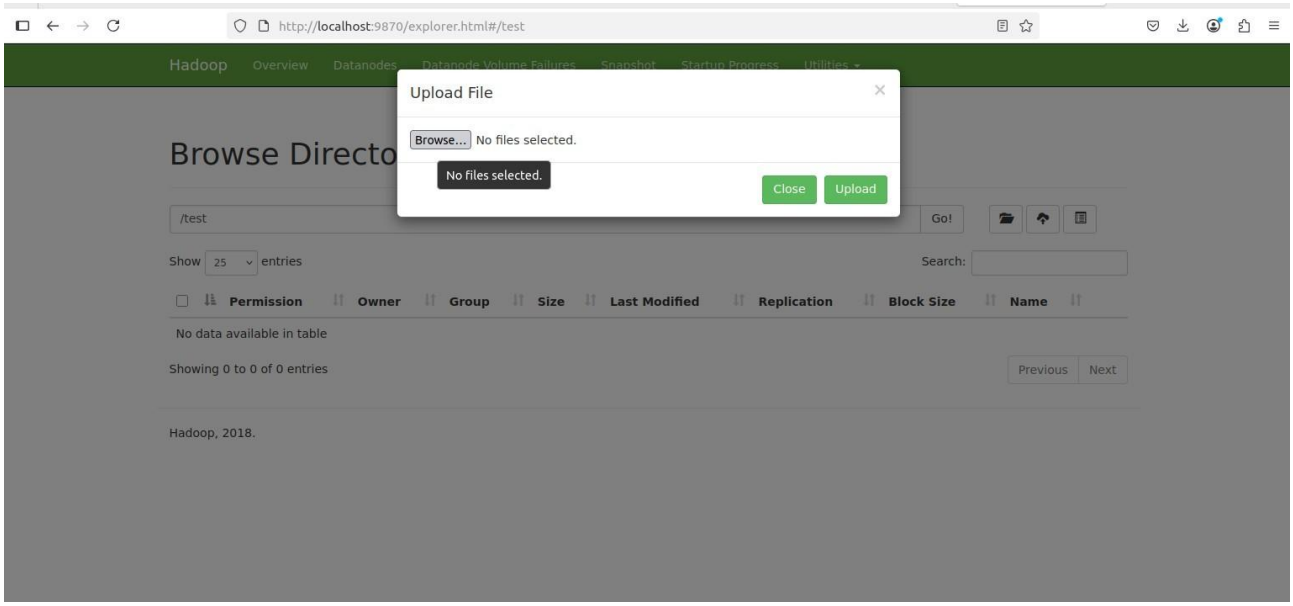
```
mohit@mohit-virtual-machine:~$
mohit@mohit-virtual-machine:~$ sudo su - hadoop
hadoop@mohit-virtual-machine:~$ start-dfs.sh
Starting namenodes on [0.0.0.0]
Starting datanodes
Starting secondary namenodes [mohit-virtual-machine]
hadoop@mohit-virtual-machine:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
hadoop@mohit-virtual-machine:~$
```

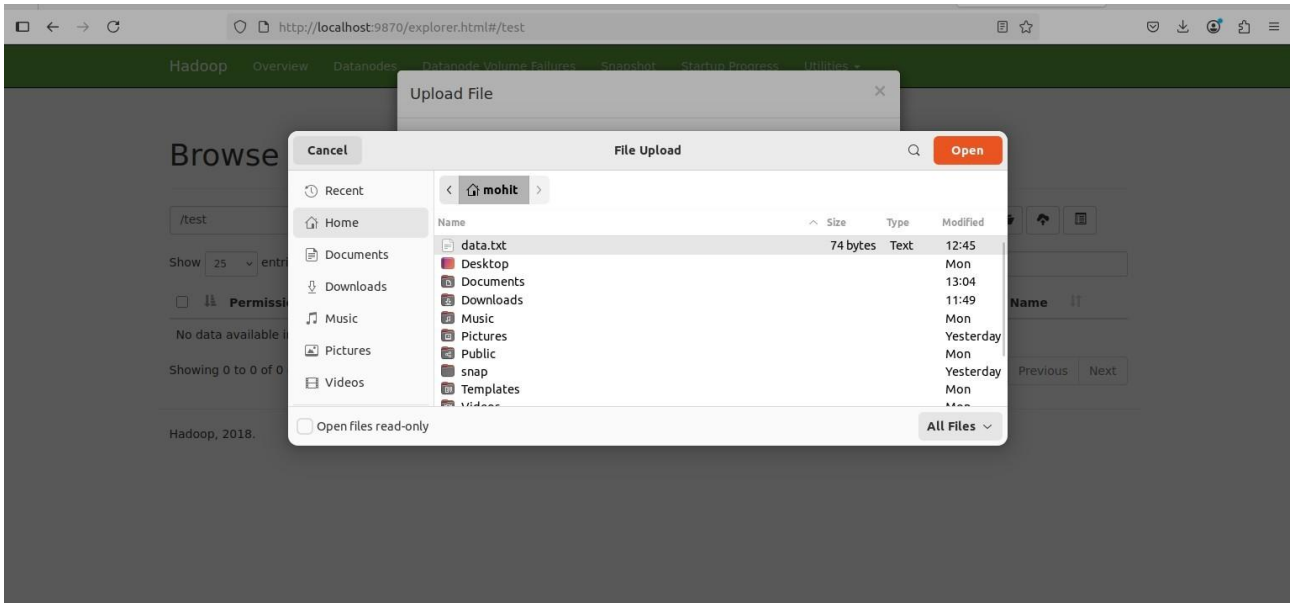
Create a folder in Hadoop and upload the data.txt file

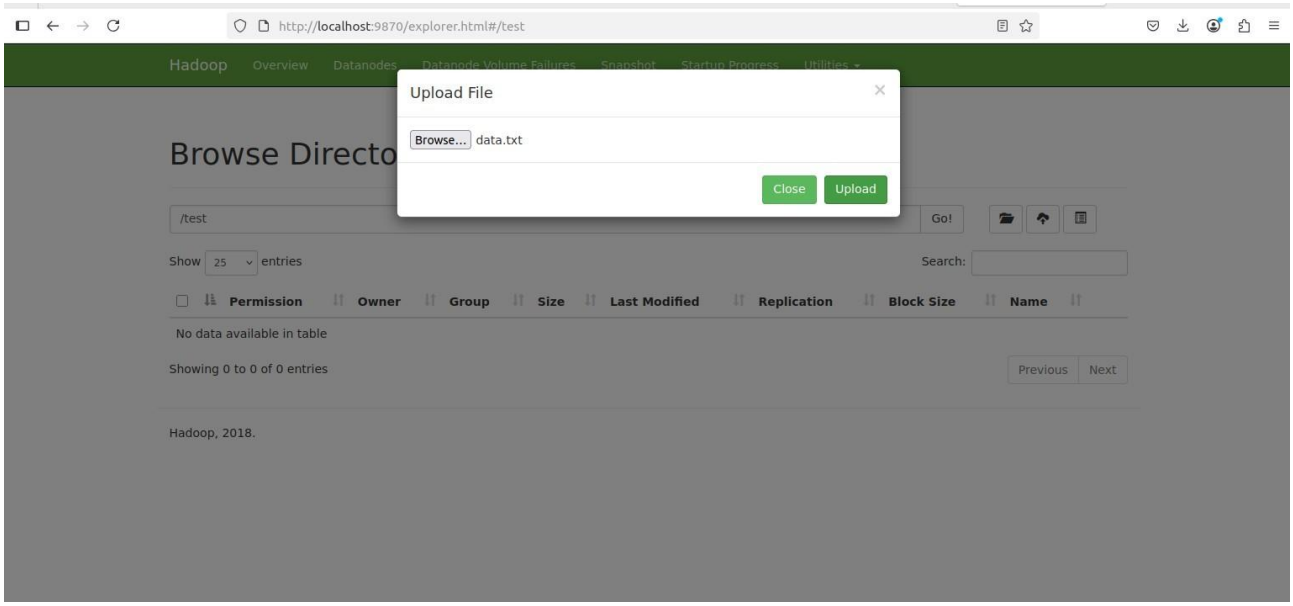

http://localhost:9870/dfshealth.html#tab-overview
Hadoop
Overview
Datanodes
Datanode Volume Failures
Snapshot
Startup Progress
Utilities
Browse the file system
Logs
Metrics
Configuration
Process Thread Dump
Overview '0.0.0.0:9000' (active)
Started: Wed Jun 11 12:55:38 +0530 2025
Version: 3.1.1, r2b9a8c1d3a2caf1e733d57f346af3ff0d5ba529c
Compiled: Thu Aug 02 09:56:00 +0530 2018 by leftnoteasy from branch-3.1.1
Cluster ID: CID-621a8db9-02fd-4cd2-b100-58f18157f3cb
Block Pool ID: BP-1751532922-127.0.1.1-1749532332875
Summary
Security is off.
Safemode is off.
1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).
Heap Memory used 84.89 MB of 239.5 MB Heap Memory. Max Heap Memory is 860.5 MB.
Non Heap Memory used 47.66 MB of 48.63 MB Commited Non Heap Memory. Max Non Heap Memory is <unbounded>.


http://localhost:9870/explorer.html#/
Hadoop
Overview
Datanodes
Create Directory
/
test
Cancel
Create
Browse Directo
Go!
Show 25 entries
Search:
Permission
Owner
Group
Size
Last Modified
Replication
Block Size
Name
No data available in table
Showing 0 to 0 of 0 entries
Previous
Next
Hadoop, 2018.


http://localhost:9870/explorer.html#/
Hadoop
Overview
Datanodes
Datanode Volume Failures
Snapshot
Startup Progress
Utilities
Browse Directory
/
Go!
Show 25 entries
Search:
Permission
Owner
Group
Size
Last Modified
Replication
Block Size
Name
drwxr-xr-x
dr.who
supergroup
0 B
Jun 11 13:03
0
0 B
test
Showing 1 to 1 of 1 entries
Previous
1
Next
Hadoop, 2018.


http://localhost:9870/explorer.html#/test
Hadoop
Overview
Datanodes
Datanode Volume Failures
Snapshot
Startup Progress
Utilities
Upload File
Browse...
No files selected.
No files selected.
Close
Upload
Browse Directo
/test
Go!
Show 25 entries
Search:
Permission
Owner
Group
Size
Last Modified
Replication
Block Size
Name
No data available in table
Showing 0 to 0 of 0 entries
Previous
Next
Hadoop, 2018.





Compliling (make sure to login as Hadoop user)
Use cd command to go to directory were wordcountproblem.jar is saved
hadoop jar ~/wordcountproblem.jar com.wordcountproblem.WC_Runner /test/data.txt /r_output

```
hadoop@mohit-virtual-machine:~$ ls
hadoop-3.1.1.tar.gz  hdfs  snap  wordcountproblem.jar
hadoop@mohit-virtual-machine:~$
hadoop@mohit-virtual-machine:~$
hadoop@mohit-virtual-machine:~$ hadoop jar ~/wordcountproblem.jar com.wordcountp
roblem.WC_Runner /test/data.txt /r_output
2025-06-11 14:14:55,042 INFO client.RMProxy: Connecting to ResourceManager at /0
.0.0.0:8032
```

Running

hdfs dfs -cat /r_output/part-00000

```
hadoop@mohit-virtual-machine:~$
hadoop@mohit-virtual-machine:~$ hdfs dfs -cat /r_output/part-00000
HDFS    1
Hadoop  1
MapReduce       1
a       2
hadoop  1
is      2
of      2
processing      1
storage 1
unit    2
hadoop@mohit-virtual-machine:~$
```

Practical No – 3

Aim: Write a program to Implement a tri-gram model

```
from collections import defaultdict
import random

def generate_trigrams(text):
"""
Builds a trigram model from input text.
A trigram is a sequence of three consecutive words.
"""
words = text.split()
trigram_model = defaultdict(list)
for i in range(len(words) - 2):
key = (words[i], words[i + 1])
next_word = words[i + 2]
trigram_model[key].append(next_word)
return trigram_model

def generate_text(model, start_words, num_words=10):
"""
Generate new text using a trigram model starting from a given word pair.
"""
if start_words not in model:
return "The given start words are not in the model."

current_words = list(start_words)
output = list(current_words)

for _ in range(num_words):
next_words = model.get(tuple(current_words))
if not next_words:
break
next_word = random.choice(next_words) output.append(next_word)
current_words = [current_words[1], next_word]

return ' '.join(output)


input_text = "I wish I may I wish I might"


model = generate_trigrams(input_text)
print ( "Tri-gram Model:" )
for key, values in model.items():
```

```
print    (f"  {key} -> {values}                    ")


print    ( "\nGenerated Text:"                    )
start = ('I', 'wish'                 )
print(generate_text(model, start, num_words=10))
```

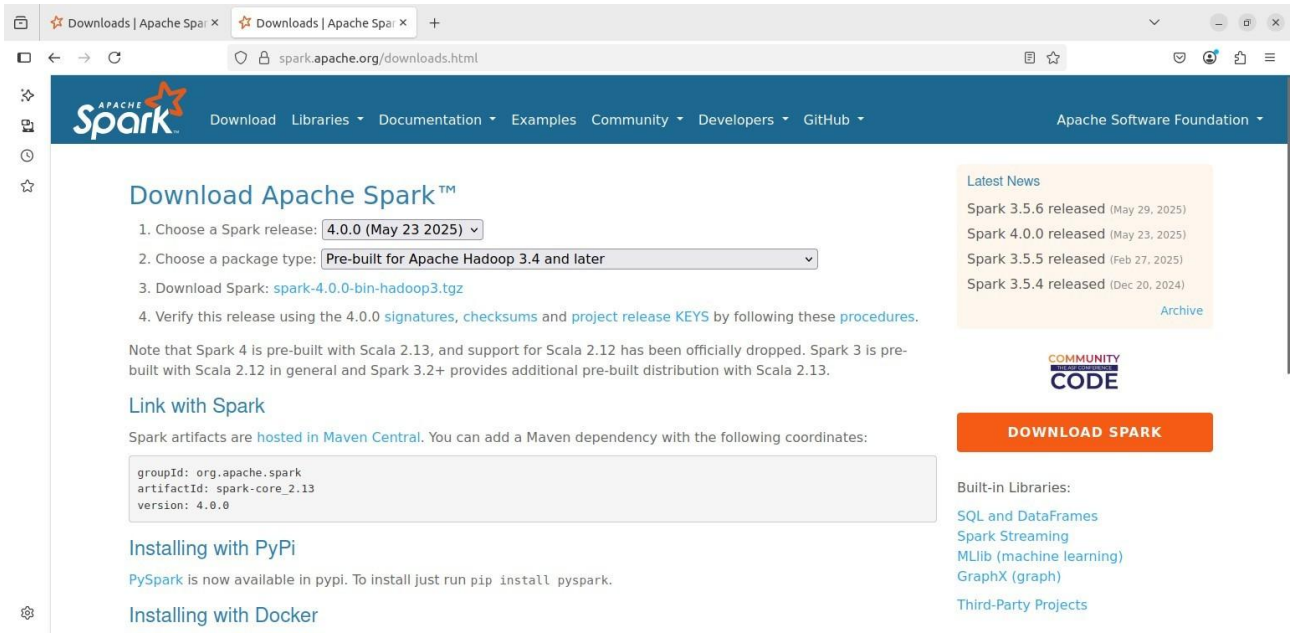
```
Tri-gram Model:
('I', 'wish') -> ['I', 'I']
('wish', 'I') -> ['may', 'might']
('I', 'may') -> ['I']
('may', 'I') -> ['wish']

Generated Text:
I wish I might
```

Practical No – 4

Aim: Download and install Spark. Create Graphical data and access the graphical data using Spark

Introduction:
Apache Spark is a powerful, open-source, distributed computing system used for big data processing and analytics. It's designed to handle both batch and real-time workloads, making it suitable for a wide range of applications, including machine learning, interactive queries, and stream processing. Spark is known for its speed and ease of use, providing a user-friendly interface for managing and executing distributed data processing tasks

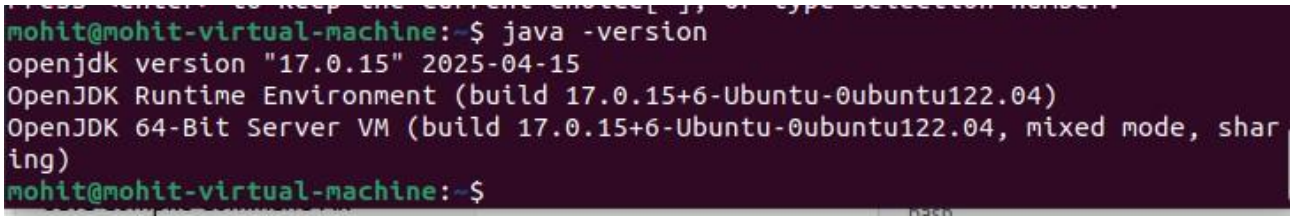
for intalltion of spart visit the given link below https://spark.apache.org/downloads.html



STEP 1: Download & Install Apache Spark

Open terminal and run: now check your java

version



1. Download Spark (with Hadoop 3)
wget https://downloads.apache.org/spark/spark-4.0.0/spark-4.0.0-bin-hadoop3.tgz
2. Extract the archive tar -xzf
spark-4.0.0-bin-hadoop3.tgz

3. Move to a simpler location mv
spark-4.0.0-bin-hadoop3 ~/spark

```
Documents  Pictures   Templates  wordcountproblem
mohit@mohit-virtual-machine:~$ cd Downloads
mohit@mohit-virtual-machine:~/Downloads$ ls
 hadoop-core-1.2.1.jar  'Practical no-1.pdf'   spark-4.0.0-bin-hadoop3.tgz
mohit@mohit-virtual-machine:~/Downloads$ tar -xzf spark-4.0.0-bin-hadoop3.tgz
mohit@mohit-virtual-machine:~/Downloads$ mv spark-4.0.0-bin-hadoop3 ~/spark
mohit@mohit-virtual-machine:~/Downloads$
```

STEP 2: Set Environment Variables (Optional but recommended)

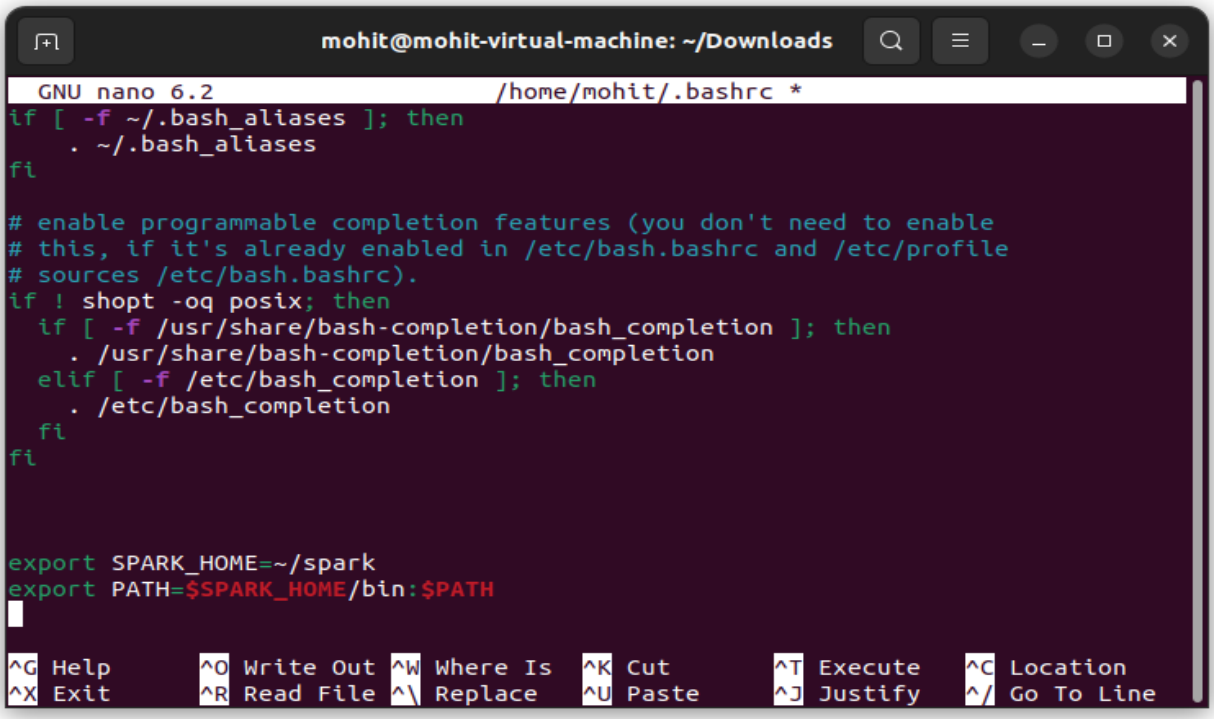
Edit your ~/.bashrc or ~/.zshrc:

nano ~/.bashrc

Add at the bottom:

export SPARK_HOME=~/spark export

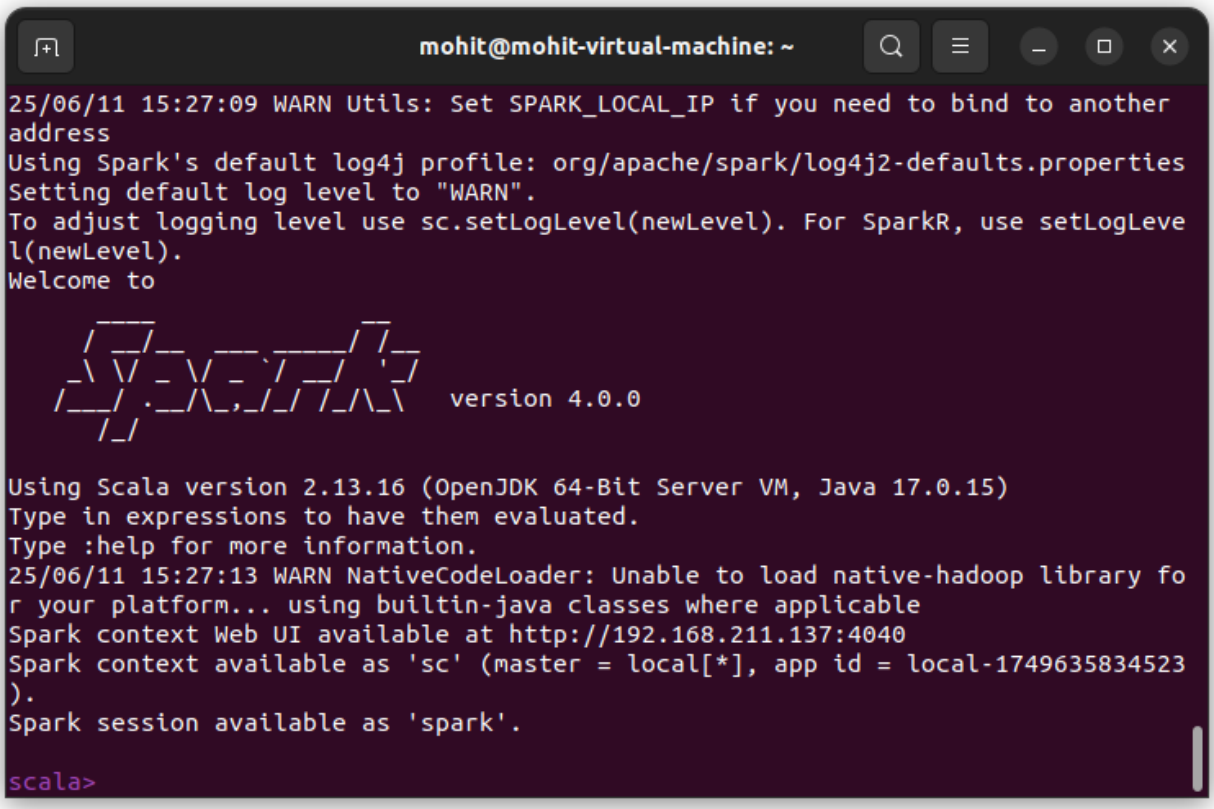
PATH=$SPARK_HOME/bin:$PATH



Then apply it:
source ~/.bashrc

```
mohit@mohit-virtual-machine:~/Downloads$ source ~/.bashrc
mohit@mohit-virtual-machine:~/Downloads$
```

STEP 3: Run spark-shell (Scala-based REPL for Spark) spark-shell



Create Graphical data and access the graphical data using Spark
Type the code line by line in the spark-shell
Type spark-shell to access the shell

Entering graphical data in spark shell

#creating graphical data in graphx
import org.apache.spark.graphx._

creating own data type case class
User(name: String, age: Int) val users
= List((1L, User("Alex", 26)), (2L,
User("Bill", 42)), (3L, User("Carol",
18)), (4L, User("Dave",16)), (5L,

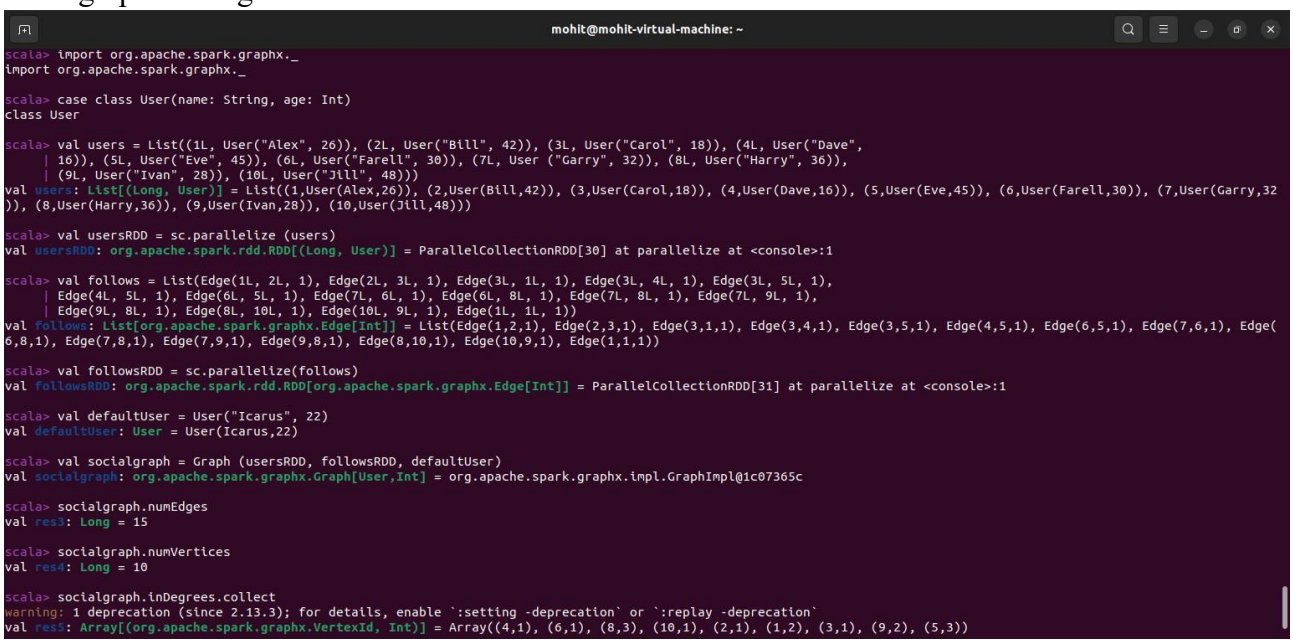
User("Eve", 45)), (6L, User("Farell",
30)), (7L, User ("Garry", 32)), (8L,
User("Harry", 36)),(9L, User("Ivan", 28)), (10L, User("Jill", 48)))

val usersRDD = sc.parallelize (users)
val follows = List(Edge(1L, 2L, 1), Edge(2L, 3L, 1), Edge(3L, 1L, 1), Edge(3L, 4L, 1), Edge(3L, 5L, 1),
Edge(4L, 5L, 1), Edge(6L, 5L, 1), Edge(7L, 6L, 1), Edge(6L, 8L, 1), Edge(7L, 8L, 1), Edge(7L, 9L, 1),Edge(9L, 8L, 1), Edge(8L, 10L, 1), Edge(10L, 9L, 1), Edge(1L, 1L, 1))

val followsRDD = sc.parallelize(follows)

# creating user to access data 19 val
defaultUser = User("Icarus", 22)
val socialgraph = Graph (usersRDD, followsRDD, defaultUser)

#Access data of the graph
socialgraph.numEdges
socialgraph.numVertices
socialgraph.inDegrees.collect
socialgraph.outDegrees.collect



press ctrl + D for exit the shell

Practical No - 5

Aim: Write a Spark code for the given application and handle error and recovery of data

Introduction:
Scala provides robust constructs for functional error handling. These include:

- Try, Success, Failure – for wrapping computations that may fail.
- Option, Some, None – for handling optional values.
- Either, Left, Right – for results that can be two possible types (usually success or failure).

Install sbt (Scala Build Tool)

sudo apt update sudo apt

install curl -y

```
mohit@mohit-virtual-machine:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  curl
```

sudo apt install gnupg -y

```
mohit@mohit-virtual-machine:~$ sudo apt update
sudo apt install gnupg -y
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
```

sudo apt-key adv --keyserver hkp://keyserver.ubuntu.com:80 \
--recv-keys 2EE0EA64E40A89B84B2DF73499E82A75642AC823

```
mohit@mohit-virtual-machine:~$ sudo apt-key adv --keyserver hkp://keyserver.ubun
tu.com:80 \
  --recv-keys 2EE0EA64E40A89B84B2DF73499E82A75642AC823
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (s
ee apt-key(8)).
Executing: /tmp/apt-key-gpghome.RQloTywQC2/gpg.1.sh --keyserver hkp://keyserver.
```

echo "deb https://repo.scala-sbt.org/scalasbt/debian all main" | \
sudo tee /etc/apt/sources.list.d/sbt.list

echo "deb https://repo.scala-sbt.org/scalasbt/debian /" | \
sudo tee /etc/apt/sources.list.d/sbt_old.list

```
mohit@mohit-virtual-machine:~$ echo "deb https://repo.scala-sbt.org/scalasbt/deb
ian all main" | \
  sudo tee /etc/apt/sources.list.d/sbt.list
echo "deb https://repo.scala-sbt.org/scalasbt/debian /" | \
  sudo tee /etc/apt/sources.list.d/sbt_old.list
deb https://repo.scala-sbt.org/scalasbt/debian all main
deb https://repo.scala-sbt.org/scalasbt/debian /
```

sudo apt update sudo
apt install sbt -y

```
mohit@mohit-virtual-machine:~$ sudo apt update
sudo apt install sbt -y
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
```

echo "deb [signed-by=/usr/share/keyrings/sbt-archive-keyring.gpg]
https://repo.scala-sbt.org/scalasbt/debian all main" | sudo tee /etc/apt/sources.list.d/sbt.list

```
mohit@mohit-virtual-machine:~$ echo "deb [signed-by=/usr/share/keyrings/sbt-arch
ive-keyring.gpg] https://repo.scala-sbt.org/scalasbt/debian all main" | sudo tee
 /etc/apt/sources.list.d/sbt.list
deb [signed-by=/usr/share/keyrings/sbt-archive-keyring.gpg] https://repo.scala-s
bt.org/scalasbt/debian all main
```

curl -sL "https://keyserver.ubuntu.com/pks/lookup?
op=get&search=0x2EE0EA64E40A89B84B2DF73499E82A75642AC823" | sudo apt-key add -

```
mohit@mohit-virtual-machine:~$ curl -sL "https://keyserver.ubuntu.com/pks/lookup
?op=get&search=0x2EE0EA64E40A89B84B2DF73499E82A75642AC823" | sudo apt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (s
ee apt-key(8)).
OK
```

sudo apt-get update

Verify sbt installation

Sbt

```
mohit@mohit-virtual-machine:~$ sbt
[error] Neither build.sbt nor a 'project' directory in the current directory: /h
ome/mohit
[error] run 'sbt new', touch build.sbt, or run 'sbt --allow-empty'.
[error]
[error] To opt out of this check, create /home/mohit/.config/sbt/sbtopts with:
[error] --allow-empty
mohit@mohit-virtual-machine:~$
```

Install scala

wget https://downloads.lightbend.com/scala/2.12.18/scala-2.12.18.deb

sudo dpkg -i scala-2.12.18.deb

```
mohit@mohit-virtual-machine:~$ scala -version
Scala code runner version 2.12.18 -- Copyright 2002-2023, LAMP/EPFL and Lightben
d, Inc.
```

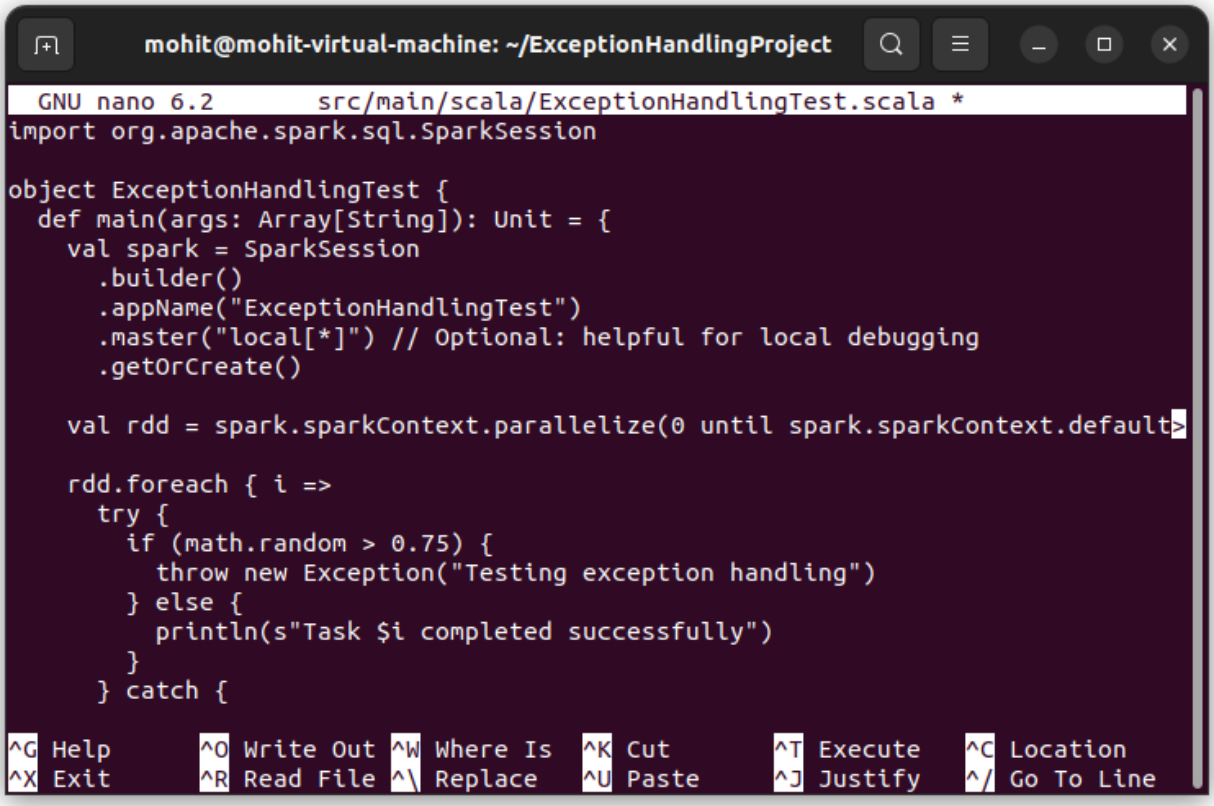
Create Scala source file nano

ExceptionHandlingTest.scala

```
mohit@mohit-virtual-machine:~$ nano ExceptionHandlingTest.scala
mohit@mohit-virtual-machine:~$
```

```
import org.apache.spark.sql.SparkSession
object ExceptionHandlingTest {  def
main(args: Array[String]): Unit = {    val
spark = SparkSession
    .builder()
    .appName("ExceptionHandlingTest")
    .master("local[*]") // Optional: helpful for local debugging
    .getOrCreate()

  val rdd = spark.sparkContext.parallelize(0 until spark.sparkContext.defaultParallelism)

  rdd.foreach { i =>
try {
     if (math.random > 0.75) {
       throw new Exception("Testing exception handling")
     } else {
       println(s"Task $i completed successfully")
```

```
      }
    } catch {        case e: Exception =>
println(s"Exception in task $i: ${e.getMessage}")
    }
}

  spark.stop()
 }
}
```



Start Spark Master and Worker

$SPARK_HOME/sbin/start-master.sh

$SPARK_HOME/sbin/start-worker.sh spark://localhost:7077

```
mohit@mohit-virtual-machine:~$ $SPARK_HOME/sbin/start-master.sh
starting org.apache.spark.deploy.master.Master, logging to /home/mohit/spark/log
s/spark-mohit-org.apache.spark.deploy.master.Master-1-mohit-virtual-machine.out
mohit@mohit-virtual-machine:~$ $SPARK_HOME/sbin/start-worker.sh spark://localhos
t:7077
starting org.apache.spark.deploy.worker.Worker, logging to /home/mohit/spark/log
s/spark-mohit-org.apache.spark.deploy.worker.Worker-1-mohit-virtual-machine.out
mohit@mohit-virtual-machine:~$
```

Build the Scala project

```
mohit@mohit-virtual-machine:~$ ls
data.txt   exceptionhandlingtest.sbt    Public     Videos
Desktop    ExceptionHandlingTest.scala  snap       wordcountclasses
Documents  Music                        spark      wordcountproblem
Downloads  Pictures                     Templates  wordcountproblem.jar
mohit@mohit-virtual-machine:~$
```

mkdir ~/ExceptionHandlingProject cd
~/ExceptionHandlingProject mkdir -p
src/main/scala mv ~/ExceptionHandlingTest.scala
src/main/scala/ nano build.sbt

```
mohit@mohit-virtual-machine:~$
mohit@mohit-virtual-machine:~$ mkdir ~/ExceptionHandlingProject
cd ~/ExceptionHandlingProject
mohit@mohit-virtual-machine:~/ExceptionHandlingProject$ mkdir -p src/main/scala
mohit@mohit-virtual-machine:~/ExceptionHandlingProject$ mv ~/ExceptionHandlingTe
st.scala src/main/scala/
mohit@mohit-virtual-machine:~/ExceptionHandlingProject$ nano build.sbt
```

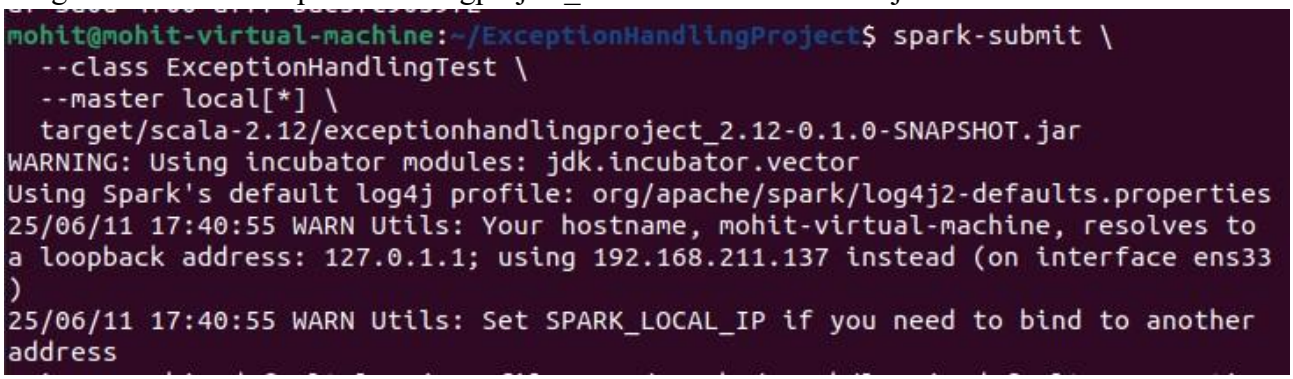
scalaVersion := "2.12.18"

libraryDependencies ++= Seq(
"org.apache.spark" %% "spark-core" % "3.4.1",
"org.apache.spark" %% "spark-sql" % "3.4.1"
)

```
mohit@mohit-virtual-machine: ~/ExceptionHandlingProject
  GNU nano 6.2                          build.sbt
scalaVersion := "2.12.18"

libraryDependencies ++= Seq(
  "org.apache.spark" %% "spark-core" % "3.4.1",
  "org.apache.spark" %% "spark-sql"  % "3.4.1"
)
```

sbt package

Submit your program to Spark spark-
submit \
  --class ExceptionHandlingTest \
  --master local[*] \
  target/scala-2.12/exceptionhandlingproject_2.12-0.1.0-SNAPSHOT.jar



Practical No – 6

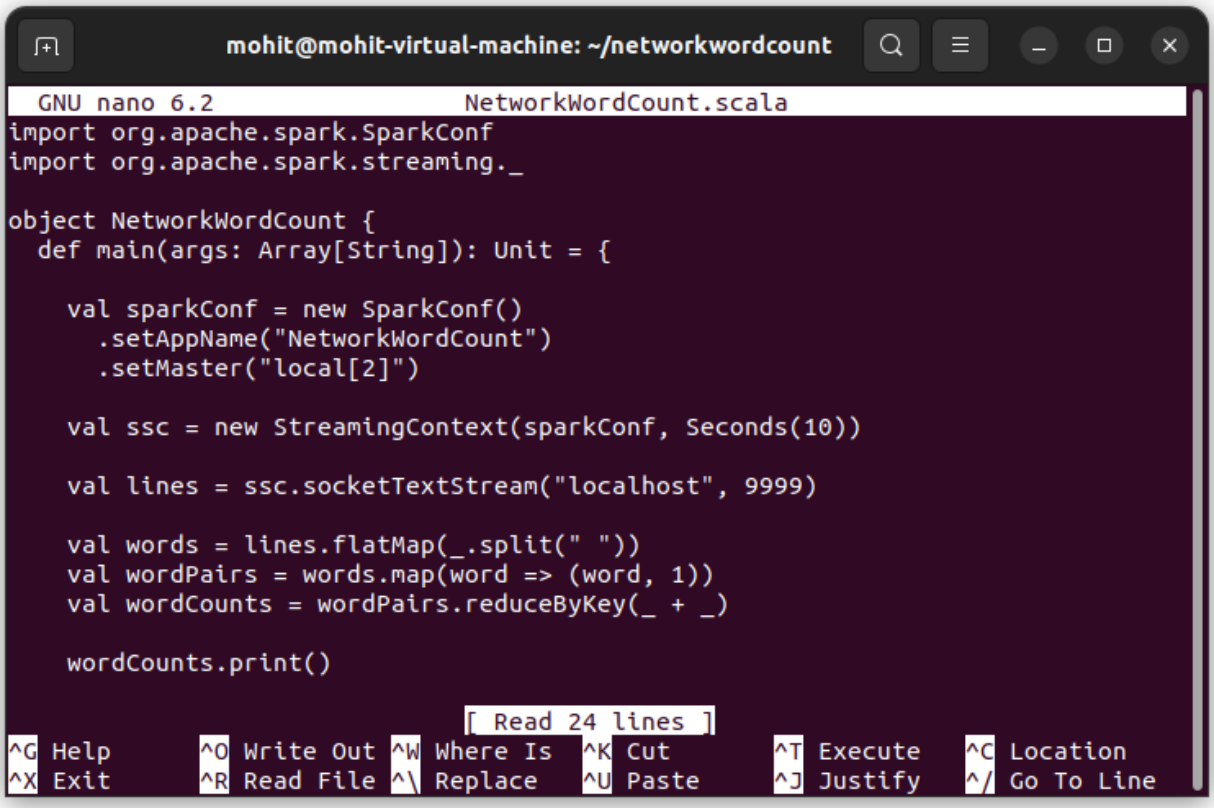
Aim: Write a Spark code to Handle the Streaming of data.

Introduction:
Spark Streaming enables scalable, high-throughput, fault-tolerant stream processing of live data streams. It divides the live data stream into **mini-batches**, performs **RDD** transformations, and outputs results in near real-time. Spark Streaming uses the same **Spark Core API**, which simplifies implementation and integration with batch and streaming workloads (Lambda Architecture).

Step 1: Project Setup (on Hadoop user)
mkdir ~/networkwordcount
cd ~/networkwordcount

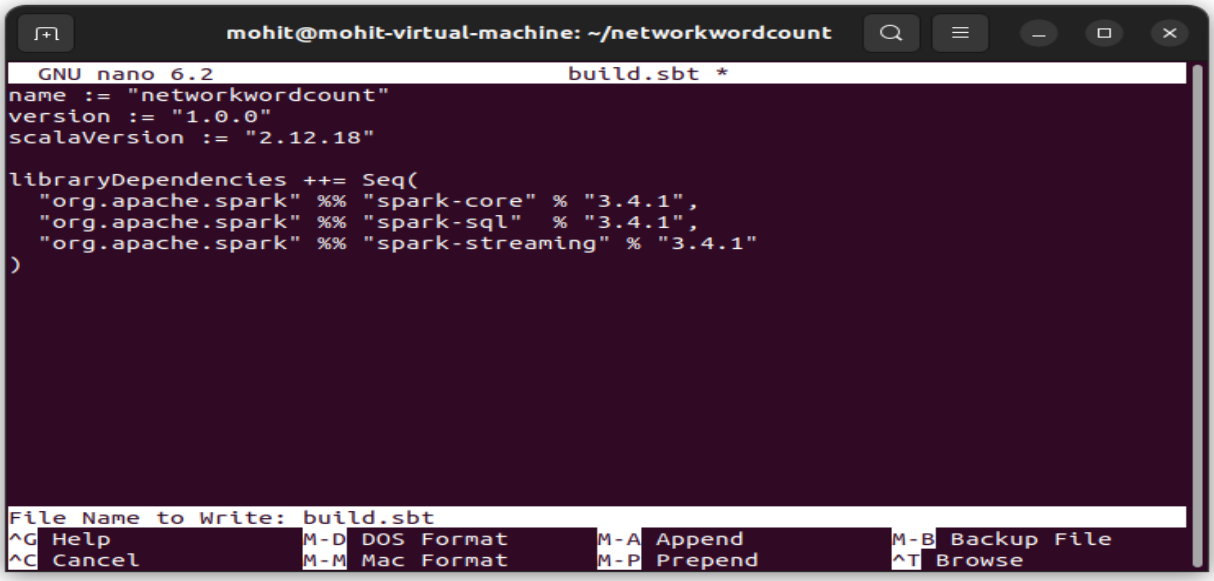
nano NetworkWordCount.scala

```
import org.apache.spark.SparkConf
import org.apache.spark.streaming._

object NetworkWordCount {   def
main(args: Array[String]): Unit = {

   val sparkConf = new SparkConf()
.setAppName("NetworkWordCount")
     .setMaster("local[2]")     val ssc = new

StreamingContext(sparkConf, Seconds(10))     val lines =

ssc.socketTextStream("localhost", 9999)

   val words = lines.flatMap(_.split(" "))     val
wordPairs = words.map(word => (word, 1))     val
wordCounts = wordPairs.reduceByKey(_ + _)
wordCounts.print()

   ssc.start()
   ssc.awaitTermination()
  }
}
```

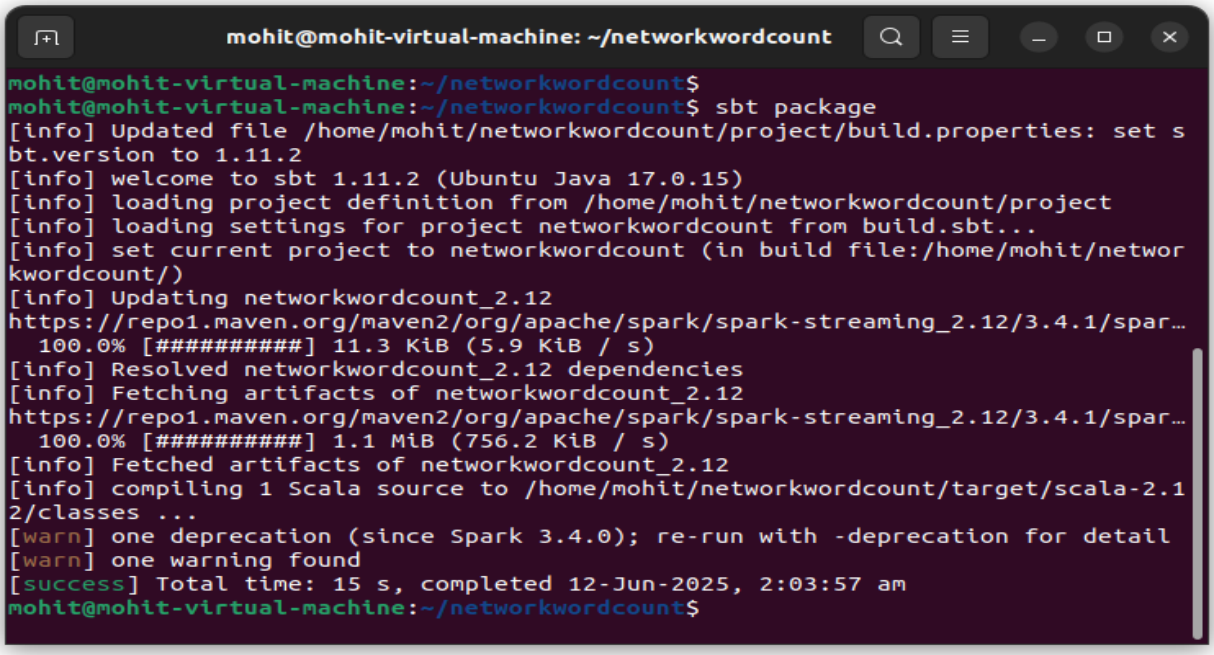


nano build.sbt name :=
"networkwordcount" version
:= "1.0.0" scalaVersion :=
"2.12.18"

libraryDependencies ++= Seq(
"org.apache.spark" %% "spark-core" % "3.4.1",
"org.apache.spark" %% "spark-sql" % "3.4.1",
"org.apache.spark" %% "spark-streaming" % "3.4.1"
)

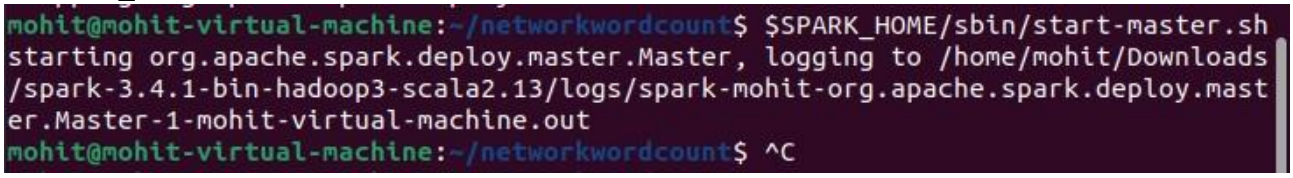


sbt package



Start the spark server

$SPARK_HOME/sbin/start-master.sh



$SPARK_HOME/sbin/start-worker.sh spark://mohit-virtual-machine:7077

```
mohit@mohit-virtual-machine:~/networkwordcount$ $SPARK_HOME/sbin/start-worker.sh
 spark://mohit-virtual-machine:7077
starting org.apache.spark.deploy.worker.Worker, logging to /home/mohit/Downloads
/spark-3.4.1-bin-hadoop3-scala2.13/logs/spark-mohit-org.apache.spark.deploy.work
er.Worker-1-mohit-virtual-machine.out
```

spark-submit \
--class NetworkWordCount \
--master spark://mohit-virtual-machine:7077 \ target/scala-2.12/networkwordcount_2.12-1.0.0.jar

```
mohit@mohit-virtual-machine:~/networkwordcount$ spark-submit --class NetworkWord
Count \
  --master spark://mohit-virtual-machine:7077 \
  target/scala-2.12/networkwordcount_2.12-1.0.0.jar
25/06/12 17:42:08 WARN Utils: Your hostname, mohit-virtual-machine resolves to a
 loopback address: 127.0.1.1; using 192.168.211.137 instead (on interface ens33)
25/06/12 17:42:08 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another
address
25/06/12 17:42:09 INFO SparkContext: Running Spark version 3.4.1
25/06/12 17:42:09 WARN NativeCodeLoader: Unable to load native-hadoop library fo
r your platform... using builtin-java classes where applicable
25/06/12 17:42:09 INFO ResourceUtils: ==========================================
====================
25/06/12 17:42:09 INFO ResourceUtils: No custom resources configured for spark.d
river.
25/06/12 17:42:09 INFO ResourceUtils: ==========================================
```

On a separate terminal, start the netscape server

```
mohit@mohit-virtual-machine:~$ nc -lk 9999

hello world
spark streaming is awesome
hello hello
```

OR

open the spark shell import

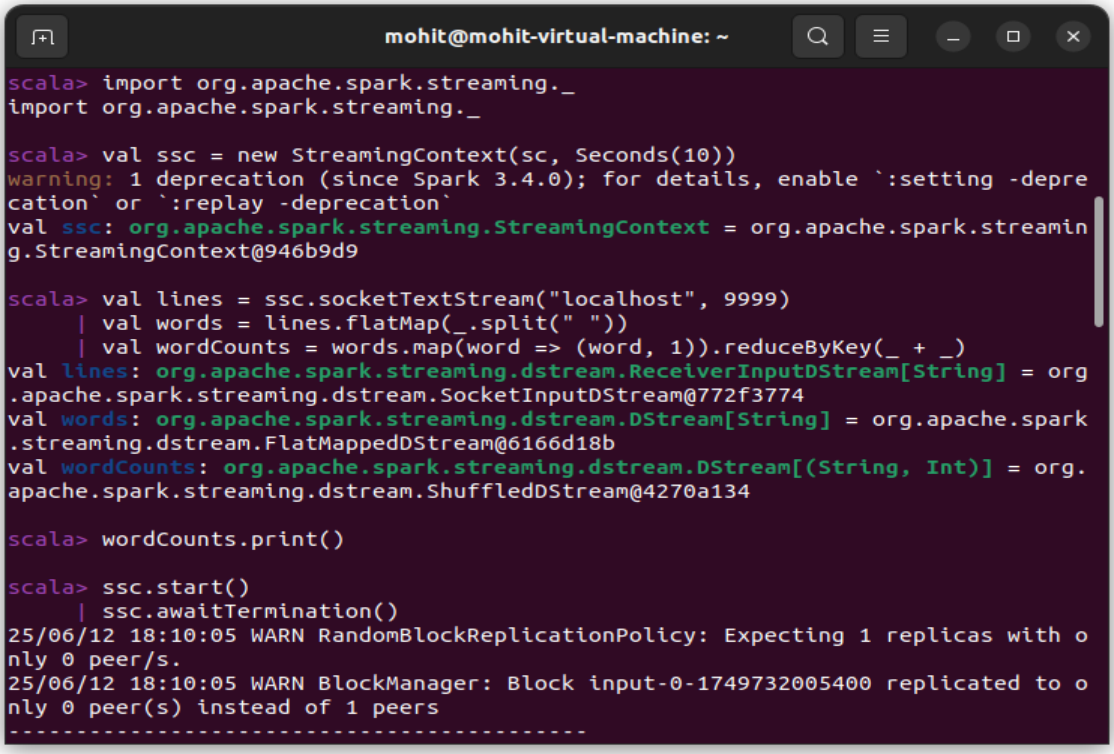
org.apache.spark.streaming._ val ssc = new

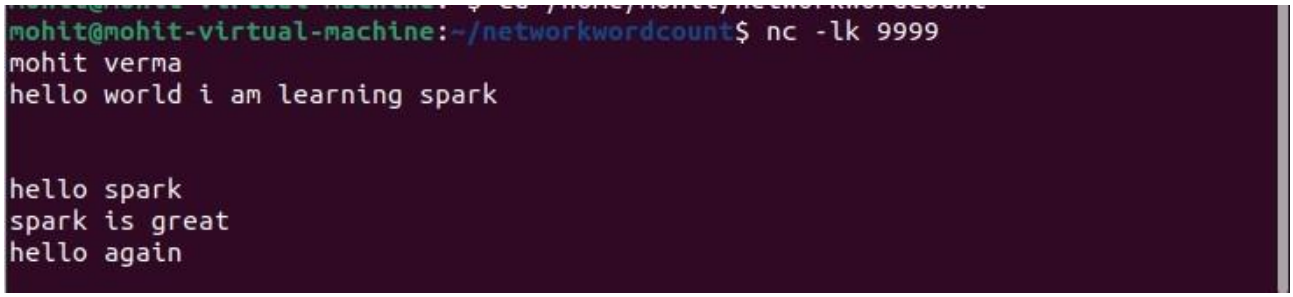
StreamingContext(sc, Seconds(5))

val lines = ssc.socketTextStream("localhost", 9999) val words = lines.flatMap(_.split(" ")) val wordCounts = words.map(word => (word, 1)).reduceByKey(_ + _) wordCounts.print()

ssc.start()

ssc.awaitTermination()





Practical No – 7

Aim: Install HBase and use the HBase Data model Store and retrieve data

Introduction:

HBase is **a column-oriented, non-relational database**. This means that data is stored in individual columns, and indexed by a unique row key. This architecture allows for rapid retrieval of individual rows and columns and efficient scans over individual columns within a table.

Strep1:

To work with HBase, we need to install Hadoop and the Java dependencies first. This was done in Practical One

start Hadoop start-

dfs.sh start-yarn.sh

```
hadoop@mohit-virtual-machine:~$ start-dfs.sh
Starting namenodes on [0.0.0.0]
0.0.0.0: namenode is running as process 8996.  Stop it first.
Starting datanodes
localhost: datanode is running as process 9264.  Stop it first.
Starting secondary namenodes [mohit-virtual-machine]
mohit-virtual-machine: secondarynamenode is running as process 9508.  Stop it fi
rst.
hadoop@mohit-virtual-machine:~$ start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 9707.  Stop it first.
Starting nodemanagers
localhost: nodemanager is running as process 9965.  Stop it first.
hadoop@mohit-virtual-machine:~$
```

Step 2:

Download Hbase

wget https://archive.apache.org/dist/hbase/2.4.1/hbase-2.4.1-bin.tar.gz

```
hadoop@mohit-virtual-machine:~$ wget https://archive.apache.org/dist/hbase/2.4.1
/hbase-2.4.1-bin.tar.gz
--2025-07-03 09:10:49--  https://archive.apache.org/dist/hbase/2.4.1/hbase-2.4.1
-bin.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a
:a084::2
```

Unzip it by executing command

tar -xvf hbase-2.4.1-bin.tar.gz

```
hadoop@mohit-virtual-machine:~$
hadoop@mohit-virtual-machine:~$ tar -xvf hbase-2.4.1-bin.tar.gz
hbase-2.4.1/LICENSE.txt
hbase-2.4.1/NOTICE.txt
hbase-2.4.1/LEGAL
hbase-2.4.1/docs/
```

Rename directory to hbase

mv hbase-2.4.1 hbase

```
hadoop@mohit-virtual-machine:~$
hadoop@mohit-virtual-machine:~$
hadoop@mohit-virtual-machine:~$ mv hbase-2.4.1 hbase
```

It will unzip the contents, and it will create hbase-2.4.15 directory in the location /home/Hadoop
Now rename the directory to hbase

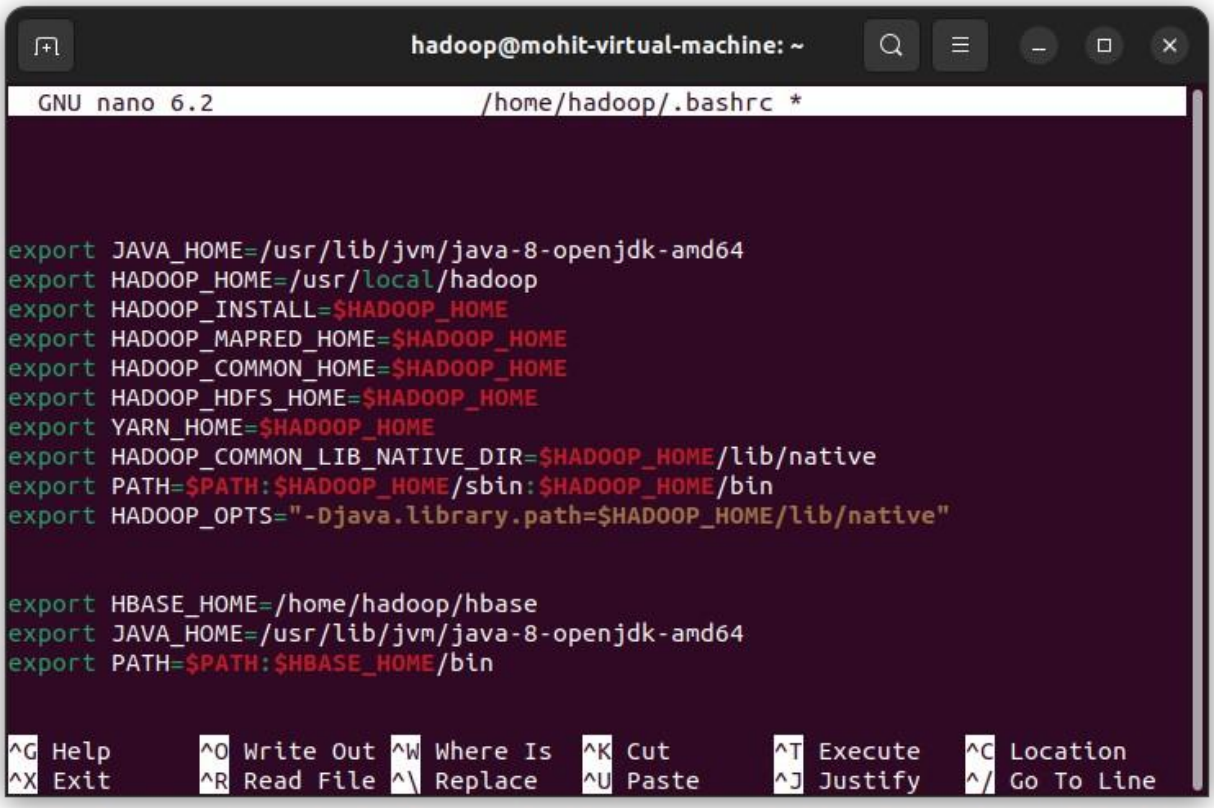
Set variables in bashc.

nano ~/.bashrc export

HBASE_HOME=/home/hadoop/hbase export

JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64 export

PATH=$PATH:$HBASE_HOME/bin



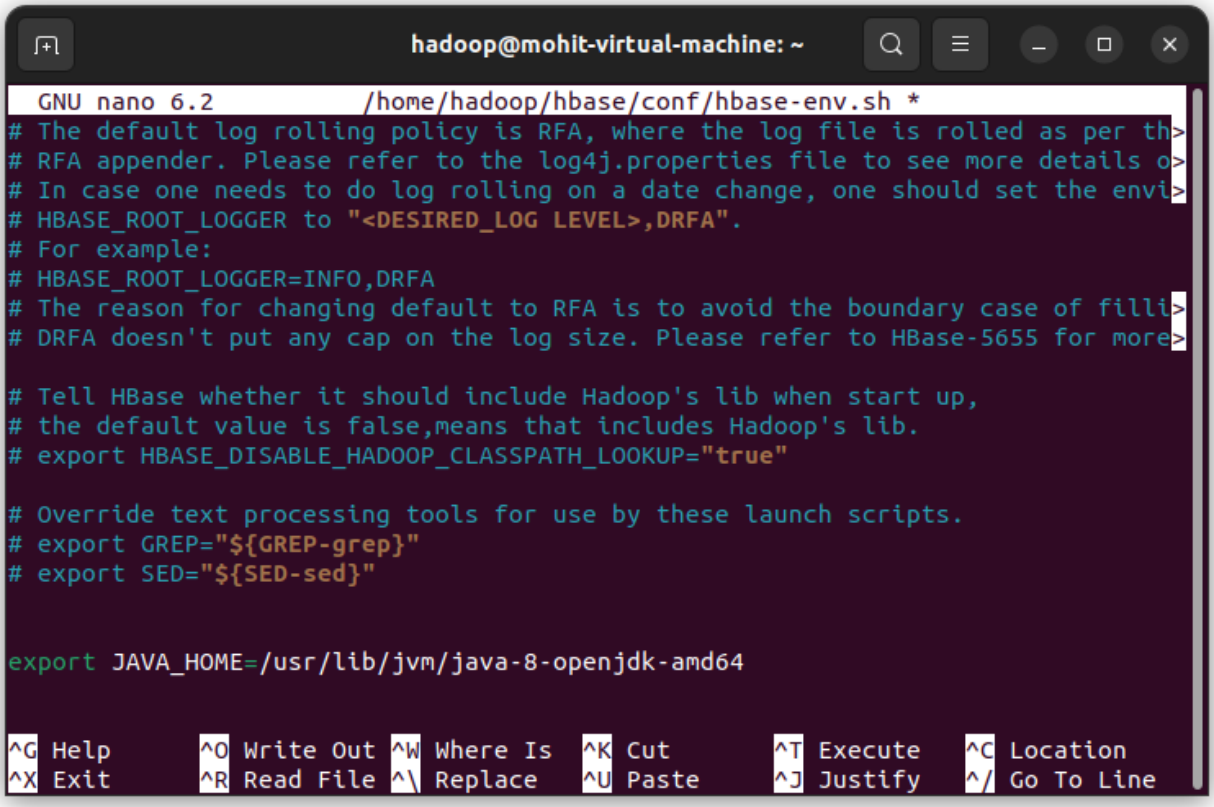
Read the edited bashrc file to the running memory source

~/.bashrc

Configure HBase:

nano ~/hbase/conf/hbase-env.sh export

JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

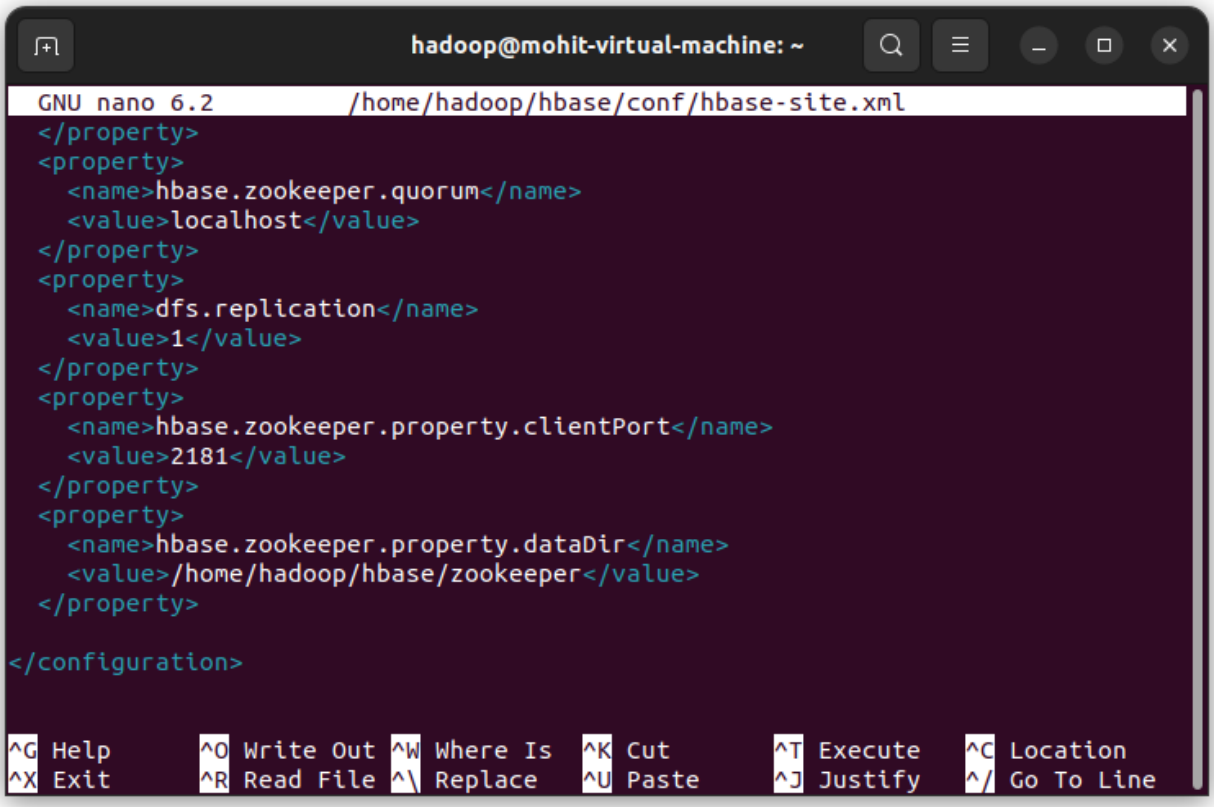


Open HBase-site.xml and mention the below properties in the file.

nano ~/hbase/conf/hbase-site.xml

```
<configuration>
   <property>
   <name>hbase.rootdir</name>
   <value>hdfs://localhost:9000/hbase</value>
  </property>
  <property>
   <name>hbase.cluster.distributed</name>
   <value>true</value>
  </property>
  <property>
   <name>hbase.zookeeper.quorum</name>
   <value>localhost</value>
  </property>
```

```
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>hbase.zookeeper.property.clientPort</name>
    <value>2181</value>
  </property>
  <property>
    <name>hbase.zookeeper.property.dataDir</name>
    <value>/home/hadoop/hbase/zookeeper</value>
  </property>
</configuration>
```



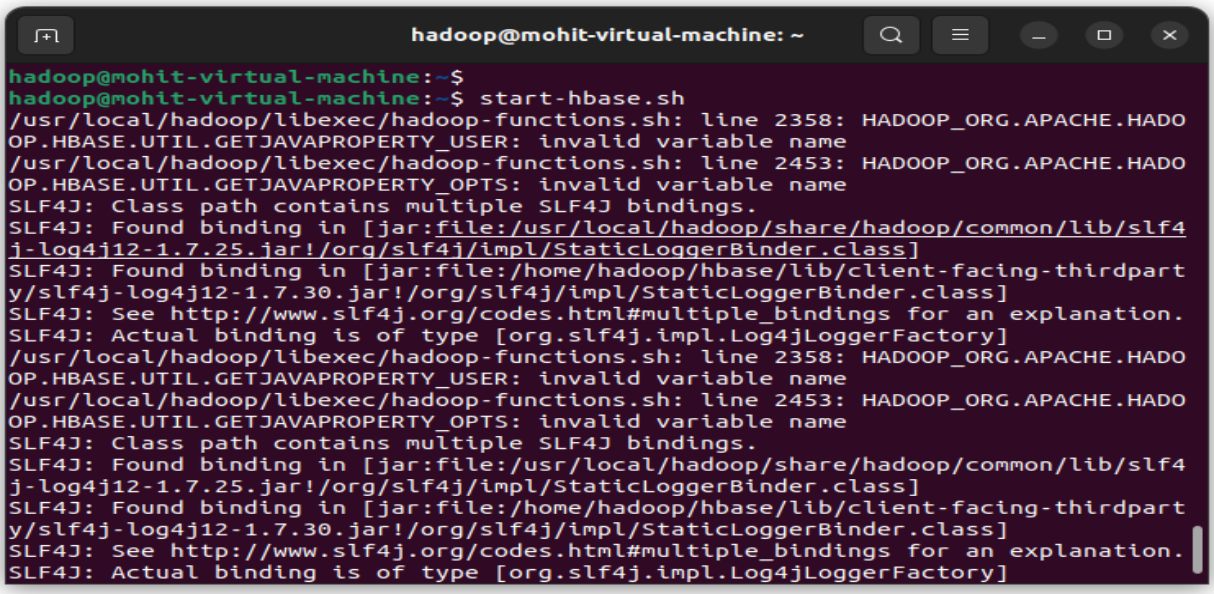
Properties Explanation

1. Setting up Hbase root directory in this property, for distributed set up we have to set thisproperty

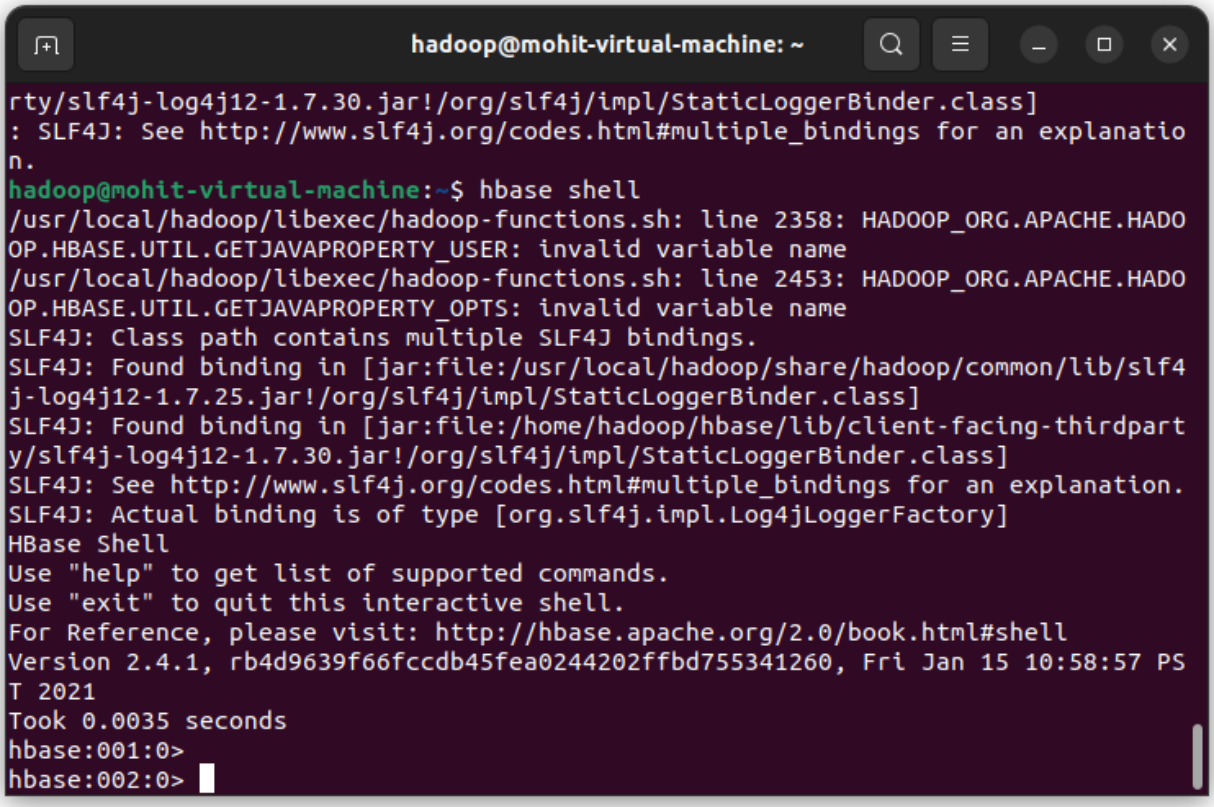
2. ZooKeeper quorum property should be set up here
3. Replication set up done in this property. By default we are placing replication as 1.In the fullydistributed mode, multiple data nodes present so we can increase replication by placing more than 1 value in the dfs.replication property 4. Client port should be mentioned in this property
5. ZooKeeper data directory can be mentioned in this property

Start Hadoop daemons first and after that start HBase daemons as shown below:

start-hbase.sh



now open Hbase shell hbase
shell



Create an HBase table and define column families

create 'emp', 'pri_data', 'pro_data'

```
Took 0.0035 seconds
hbase:001:0>
hbase:002:0> create 'emp', 'pri_data', 'pro_data'

Created table emp
Took 1.6122 seconds
=> Hbase::Table - emp
hbase:003:0>
hbase:004:0>
```

Insert Sample Data (PUT)

put 'emp', '1', 'pri_data:name', 'Andy' put
'emp', '1', 'pri_data:age', '22' put 'emp', '1',
'pro_data:post', 'asst. manager' put 'emp', '1',
'pro_data:salary', '40k' put 'emp', '2',
'pri_data:name', 'Icarus'

put 'emp', '2', 'pri_data:age', '22' put

'emp', '2', 'pro_data:post', 'manager'

```
hbase:004:0>
hbase:005:0> put 'emp', '1', 'pri_data:name', 'Andy'
Took 0.1395 seconds
hbase:006:0> put 'emp', '1', 'pri_data:age', '22'
Took 0.0070 seconds
hbase:007:0> put 'emp', '1', 'pro_data:post', 'asst. manager'
Took 0.0169 seconds
hbase:008:0> put 'emp', '1', 'pro_data:salary', '40k'
Took 0.0045 seconds
hbase:009:0> put 'emp', '2', 'pri_data:name', 'Icarus'
Took 0.0054 seconds
hbase:010:0> put 'emp', '2', 'pri_data:age', '22'
Took 0.0076 seconds
hbase:011:0> put 'emp', '2', 'pro_data:post', 'manager'
Took 0.0075 seconds
hbase:012:0>
```

Retrieve Data (GET)

get 'emp', '1'

get 'emp', '2'

```
hbase:013:0> get 'emp', '1'
COLUMN                CELL
 pri_data:age         timestamp=2025-07-03T09:40:11.438, value=22
 pri_data:name        timestamp=2025-07-03T09:40:11.396, value=Andy
 pro_data:post        timestamp=2025-07-03T09:40:11.478, value=asst. manager
 pro_data:salary      timestamp=2025-07-03T09:40:11.513, value=40k
1 row(s)
Took 0.0594 seconds
hbase:014:0> get 'emp', '2'
COLUMN                CELL
 pri_data:age         timestamp=2025-07-03T09:40:11.567, value=22
 pri_data:name        timestamp=2025-07-03T09:40:11.540, value=Icarus
 pro_data:post        timestamp=2025-07-03T09:40:11.595, value=manager
1 row(s)
Took 0.0059 seconds
hbase:015:0>
```

Delete a Specific Cell (column-level delete)

delete 'emp', '1', 'pri_data:city'

```
hbase:015:0>
hbase:016:0> delete 'emp', '1', 'pri_data:city'
Took 0.0164 seconds
hbase:017:0>
```

put 'emp', '1', 'pri_data:city', 'Delhi'

```
hbase:018:0>
hbase:019:0> put 'emp', '1', 'pri_data:city', 'Delhi'
Took 0.0057 seconds
hbase:020:0>
```

Delete Entire Row (row-level delete)

deleteall 'emp', '1'

```
hbase:020:0>
hbase:021:0> deleteall 'emp', '1'
Took 0.0115 seconds
hbase:022:0>
```

Scan All Rows scan

'emp'

```
hbase:022:0>
hbase:023:0> scan 'emp'
ROW                     COLUMN+CELL
 2                      column=pri_data:age, timestamp=2025-07-03T09:40:11.567, va
                        lue=22
 2                      column=pri_data:name, timestamp=2025-07-03T09:40:11.540, v
                        alue=Icarus
 2                      column=pro_data:post, timestamp=2025-07-03T09:40:11.595, v
                        alue=manager
1 row(s)
Took 0.0175 seconds
hbase:024:0>
```

List All Tables

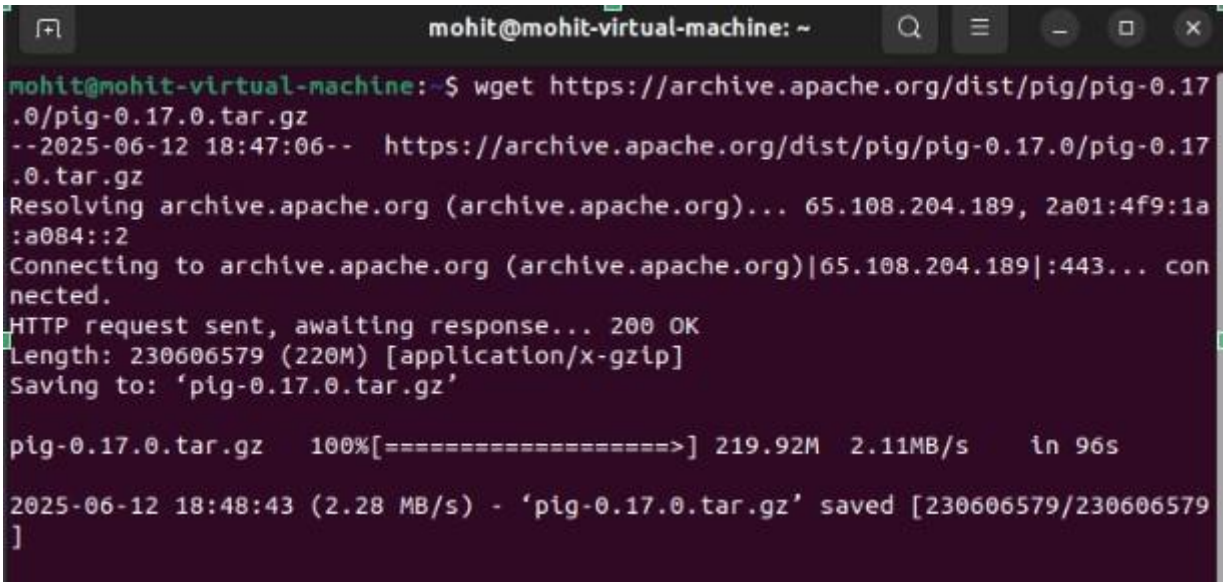
list

```
hbase:027:0>
hbase:028:0> list
TABLE
emp
1 row(s)
Took 0.0189 seconds
=> ["emp"]
hbase:029:0>
```

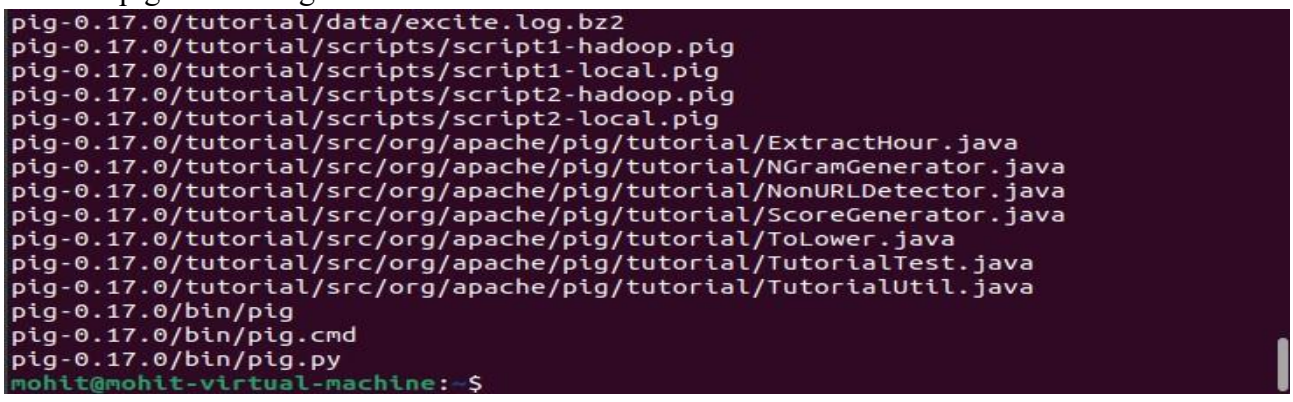
Practical No – 8

Aim: Write a Pig Script for solving counting problems.

Introducation:
Apache Pig is a high-level platform for creating programs that run on Apache Hadoop. The language for this platform is called Pig Latin. Pig can execute its Hadoop jobs in MapReduce, Apache Tez, or Apache Spark. Pig Latin abstracts the programming from the Java MapReduce idiom into a notation which makes MapReduce programming high level, similar to that of SQL for relational database management systems. wget https://downloads.apache.org/pig/pig-0.17.0/pig-0.17.0.tar.gz



tar -xvzf pig-0.17.0.tar.gz



move your hadoop and set your java_home and pig enviorment
sudo mv /home/hadoop/pig ~/ export
PIG_HOME=/home/mohit/pig
export PATH=$PATH:$PIG_HOME/bin export
PIG_CLASSPATH=$HADOOP_HOME/etc/hadoop

export JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64
export PATH=$JAVA_HOME/bin:$PATH

```
mohit@mohit-virtual-machine:~$ sudo mv /home/hadoop/pig ~/
mohit@mohit-virtual-machine:~$ nano ~/.bashrc
mohit@mohit-virtual-machine:~$ source ~/.bashrc
```

```
export SPARK_HOME=~/Downloads/spark-3.4.1-bin-hadoop3-scala2.13
export PATH=$SPARK_HOME/bin:$PATH



export PIG_HOME=/home/mohit/pig
export PATH=$PATH:$PIG_HOME/bin
export PIG_CLASSPATH=$HADOOP_HOME/etc/hadoop

export JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64
export PATH=$JAVA_HOME/bin:$PATH


^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

check the pig -version
pig -version

```
mohit@mohit-virtual-machine:~$ pig -version
Apache Pig version 0.17.0 (r1797386)
compiled Jun 02 2017, 15:41:58
```

solving counting problems
create a txt file nano
/home/mohit/textfile.txt
hello world hello hadoop pig
is powerful

```
mohit@mohit-virtual-machine: ~
  GNU nano 6.2                  /home/mohit/textfile.txt
hello world
hello hadoop
pig is powerful
```

Screenshot captured
You can paste the ima

open pig in local model
pig -x local
run this script for wordcount problem

lines = LOAD '/home/mohit/textfile.txt' AS (line:chararray); words =
FOREACH lines GENERATE FLATTEN(TOKENIZE(line)) as word;
grouped = GROUP words BY word;
wordcount = FOREACH grouped GENERATE group, COUNT(words);
DUMP wordcount;